CASTOR AND POLLUX I: THE LOG-SINE TRANSFORM G AND THE GRAM MATRIX OF THE FACTORS OF $\prod_k 2\sin(\pi x/k)$

VÍCTOR CASAÑA-GINER

ABSTRACT. Let $h(u) = \log|2\sin(\pi u)|$, so that $\sum_{k\le K} h(x/k)$ is the logarithm of the product $\prod_{k\le K}|2\sin(\pi x/k)|$. The first part of this note studies the transform $G(s) = \int_1^\infty h(u)u^{-s-1}\,du$, which is the local factor of the Mellin transform of every weighted sum of the functions $h(x/k)$. Two primitives of the Fourier series of h give the exact factorization $G(s) = -(s+1)(s+2)H(s)/(4\pi^2)$, where H is the Mellin transform of a strictly positive periodic kernel. It follows that G is entire, that on the real axis G is positive on $(-2,-1)$, vanishes simply at -1 and is negative for $s > -1$, and that a complex zero $\sigma + it$ with $\sigma > -2$ satisfies $t^2V(\sigma) \ge 2$, where V is the variance of an explicit probability measure. We also prove an asymptotic expansion of G in right sectors, which excludes zeros there, and explicit truncation and quadrature bounds with a Rouché test that isolates a zero in a disc. The second part computes the Gram matrix of the factors $x \mapsto h(x/k)$. For the mean over a common period it is the gcd matrix $\frac{\pi^2}{12}\gcd(j,k)^2/(jk)$, which gives a sum of squares with weights given by the Jordan totient J_2. For the weight $x^{-2\sigma-1}$ on $[\max(j,k),\infty)$ we prove an exact formula in which the pairs of harmonics (m,n) with $mk - nj = r$ enter through an explicit kernel, and their total weight is evaluated by the digamma function. We then study the difference $G - A$, where A is an explicit auxiliary function, using only h. On the strip $0 < \operatorname{Re} s < 2$, $G - A$ is minus the Mellin transform of the positive function $\log(\pi u/|\sin(\pi u)|)$, and, after the factor $-(s+1)(s+2)/(4\pi^2)$, the Mellin transform of an explicit positive convex kernel that satisfies a strict inversion inequality. This gives the bound $|G - A| \le A(\operatorname{Re} s) - G(\operatorname{Re} s)$, a decomposition through the digamma function, a region without solutions of $G = A$ around the real axis, and the decay of the transform. We prove analytically that $A(s) \ne 0$ for $0 \le \operatorname{Re} s \le 1$ and $|\operatorname{Im} s| \ge 9$. A positive periodic kernel with two harmonics shows that positivity and periodicity alone do not place the solutions of $G = A$ on $\operatorname{Re} s = \frac{1}{2}$. Finally, through the functional equation of the Riemann zeta function, $G(s) - A(s) = \frac{\pi}{s}\cot(\frac{\pi s}{2})\zeta(s)$, so that the statement that every non-real solution of $G(s) = A(s)$ has real part $\frac{1}{2}$ is equivalent to the Riemann hypothesis. The proofs use the Fourier series of h, Parseval's identity and integration by parts, together with the Cauchy–Schwarz inequality, Rouché's theorem and standard real and complex analysis, and, in one section only, cited results on the zeta and gamma functions. Twelve figures show the kernels, the moduli of G, of an auxiliary function A and of the Riemann zeta function on a rectangle, exponential sums over computed zeros of G and of A, and the two Gram matrices.

CONTENTS

1. Introduction 2
2. Notation 3
3. The kernel h and the transform G 5
4. The positive periodic kernel 6
5. Factorization and analytic continuation 7
6. The real axis 8
7. The singularity at $u = 1$ and a completely monotone remainder 8
8. Moments, variance and a zero-free region 9
9. Expansion in right sectors 9

2020 *Mathematics Subject Classification.* 11A05; 11C20; 42A16; 33B15; 30C15; 44A15; 11M06; 11M26.

Key words and phrases. products of sines; log-sine function; Mellin transform; positive kernel; zeros of entire functions; gcd matrices; Jordan totient; Gram matrix; digamma function; Riemann hypothesis.

10. Truncation, quadrature and a Rouché test 10
11. The mean over a period: a gcd matrix 12
12. The power weight on a half-line 13
13. Numerical comparison 16
14. Figures 17
15. The difference $G - A$ 29
16. A positive convex kernel for $G - A$ 31
17. The difference $G - A$ and the Riemann zeta function 34
18. Conclusions 35
References 37

1. Introduction

For real $u \notin \mathbb{Z}$ put

$$h(u) = \log|2\sin(\pi u)|. \tag{1.1}$$

For an integer $K \geq 1$ the sum $\sum_{k=1}^{K} h(x/k)$ is the logarithm of the product $\prod_{k=1}^{K} |2\sin(\pi x/k)|$. This note has two parts. Both use only the function h.

The first part, Sections 3 to 10, studies the transform

$$G(s) = \int_1^\infty h(u)\, u^{-s-1}\, du \qquad (\operatorname{Re} s > 0). \tag{1.2}$$

Proposition 3.2 recalls why G is the relevant object: the Mellin transform of any weighted sum $\sum_{k\leq x} \xi(k)h(x/k)$ is $G(s)$ times the Dirichlet series of the weights. Theorem 5.1 writes $G(s) = -(s+1)(s+2)H(s)/(4\pi^2)$, where H is the Mellin transform of the positive periodic kernel $\mathcal{K}(x) = \sum_{m\geq 1}(1-\cos(2\pi m x))/m^3$. Theorem 5.2 shows that G is entire and gives its values at $-1, -2, -3, -4$. Theorem 6.1 determines the sign of G on the real axis for $s > -2$. Proposition 7.1 separates the singularity of h at $u = 1$ and leaves a remainder that is a completely monotone function. Theorem 8.2 shows that a zero $\sigma + it$ of G with $\sigma > -2$, other than -1, satisfies $t^2 V(\sigma) \geq 2$, where $V(\sigma)$ is the variance of an explicit probability measure. Theorem 9.1 gives an asymptotic expansion of G in sectors $|\arg s| \leq \frac{\pi}{2} - \delta$ and excludes zeros there for $|s|$ large. Section 10 gives explicit bounds for a truncated quadrature of H and a Rouché test for a disc.

The second part, Sections 11 to 13, computes the inner products of the functions $x \mapsto h(x/j)$ and $x \mapsto h(x/k)$ for integers $j, k \geq 1$. For the mean over a common period, Theorem 11.1 shows that the Gram matrix is the gcd matrix $\frac{\pi^2}{12}\gcd(j,k)^2/(jk)$. Corollary 11.2 writes the mean square of any finite combination of factors as a sum of squares with weights given by the Jordan totient J_2, and Corollary 11.3 bounds the mean square of the first K factors between two explicit multiples of K. The mechanism of Theorem 11.1, Parseval's identity for dilates of one periodic function, is classical, and the factorization of a gcd matrix through a totient goes back to H. J. S. Smith [2]. Both are proved here in full for the kernel h. For the weight $x^{-2\sigma-1}$ on the half-line $x \geq \max(j,k)$, Theorem 12.6 gives the Gram entry exactly, as the gcd term plus an absolutely convergent series indexed by the integers $r \geq 1$. Lemma 12.2 evaluates the sum over the pairs of harmonics with $mk - nj = r$ by the digamma function, and Proposition 12.5 bounds the difference between the weighted entry and its gcd term.

Section 14 contains the figures, the continuation and the real zeros of the auxiliary function A (Proposition 14.1), and exponential sums over computed zeros of G and of A. Section 15 writes $G - A$ as minus the Mellin transform of the positive function $\log(\pi u/|\sin(\pi u)|)$ (Proposition 15.1) and through the digamma function (Proposition 15.2), and proves that $A(s) \neq 0$ for $0 \leq \operatorname{Re} s \leq 1$ and $|\operatorname{Im} s| \geq 9$ (Theorem 15.4). Section 16 writes $G - A$ through the Mellin transform of a positive convex kernel (Proposition 16.1), proves an inversion inequality for it (Proposition 16.2),

a region without solutions of $G = A$ around the real axis (Theorem 16.3) and a decay bound (Proposition 16.4), and gives a positive kernel with two harmonics for which the counterpart of $G - A$ vanishes at points with real part $\frac{3}{4}$ (Proposition 16.5). These two sections do not use the Riemann zeta function. Section 17 introduces it and proves that $G(s) - A(s) = \frac{\pi}{s}\cot(\frac{\pi s}{2})\zeta(s)$ (Proposition 17.1), so that the condition that every non-real solution of $G = A$ has real part $\frac{1}{2}$ is the Riemann hypothesis (Corollary 17.2). Section 18 states the conclusions: what is characterized without ζ, why the introduction of ζ does not lead further, and whether and where G and A intersect. Section 2 lists every symbol of the note with its single meaning.

2. Notation

Throughout, $s = \sigma + it$ is a complex number with real part σ and imaginary part t. When no s is present, σ is a real number. The letter i always denotes the imaginary unit. The letters b, j, k, l, m, n, d, r denote integers. The letters $u, x, y, v, \tau, \lambda, \theta$ denote real variables. For integers $j, k \geq 1$, (j, k) is the greatest common divisor, $[j, k]$ the least common multiple and $j \vee k = \max(j, k)$. For real x, $\{x\} = x - \lfloor x \rfloor$. Euler's constant is γ. If Φ is a locally integrable function on $\mathbb{R}$ with a period $L > 0$, its mean is

$$\operatorname{av}[\Phi] = \frac{1}{L}\int_0^L \Phi(x)\,dx, \tag{2.1}$$

which does not depend on the choice of the period L, and for an integer $\mathfrak{q}$ its Fourier coefficient with respect to the period L is $\widehat{\Phi}_L(\mathfrak{q}) = \operatorname{av}[\Phi(x)e^{-2\pi i\mathfrak{q}x/L}]$. Parseval's identity for two complex functions Φ_1, Φ_2 with period L that are square integrable on $[0, L]$ reads

$$\operatorname{av}\big[\Phi_1\overline{\Phi_2}\big] = \sum_{\mathfrak{q}\in\mathbb{Z}} \widehat{(\Phi_1)}_L(\mathfrak{q})\,\overline{\widehat{(\Phi_2)}_L(\mathfrak{q})}, \tag{2.2}$$

and the series converges absolutely. We shall use the following consequence. If a continuous function with period L has $\widehat{\Phi}_L(\mathfrak{q}) = 0$ for all $\mathfrak{q} \neq 0$, then by (2.2) with $\Phi_1 = \Phi_2 = \Phi - \widehat{\Phi}_L(0)$ it is constant.

The digamma function is defined for complex s with $\operatorname{Re} s > 0$ by

$$\psi(s) = -\gamma + \sum_{l\geq 0}\Big(\frac{1}{l+1} - \frac{1}{l+s}\Big). \tag{2.3}$$

The series converges because its terms are $O(l^{-2})$. It gives $\psi(s+1) - \psi(s) = \sum_{l\geq 0}(1/(l+s) - 1/(l+s+1)) = 1/s$, and for real $\alpha, \beta > 0$ it gives $\psi(\alpha) - \psi(\beta) = \sum_{l\geq 0}(1/(l+\beta) - 1/(l+\alpha))$.

Table 1 lists the symbols.

Table 1: Symbols, each with one meaning in the whole note.

Symbol	Meaning	Defined in
$h(u)$	$\log\lvert 2\sin(\pi u)\rvert$	(1.1)
$G(s)$	transform of h on $[1, \infty)$	(1.2)
s, σ, t	complex variable, its real and imaginary parts	Section 2
i	imaginary unit	Section 2
γ	Euler's constant	Section 2
ψ	digamma function	(2.3)
$\operatorname{av}[\Phi]$	mean of a periodic function Φ	(2.1)
$\widehat{\Phi}_L(\mathfrak{q})$	Fourier coefficient at the integer $\mathfrak{q}$, period L	Section 2
$\mathfrak{q}, \mathfrak{a}$	integer frequency indices	Section 2, (12.9)
$(j, k), [j, k], j \vee k$	gcd, lcm and maximum of j, k	Section 2

Symbol	Meaning	Defined in
$\varrho, \phi_\varrho, \phi$	Abel parameter in $(0,1)$ and the two series of Lemma 3.1	(3.3)
$\xi(k), S_\xi(x)$	weights and weighted sum of factors	(3.4)
$\mathcal{K}(x)$	positive periodic kernel	(4.1)
Σ_3, Σ_5	$\sum_{m\geq 1} m^{-3}$ and $\sum_{m\geq 1} m^{-5}$	(4.1), (5.6)
$H(s)$	Mellin transform of $\mathcal{K}$	(5.1)
$\mathcal{L}_\varpi$	weighted Gauss operator with parameter ϖ	(5.4)
$C_m(s)$	cosine integral $\int_1^\infty \cos(2\pi m u) u^{-s-3} du$	(5.7)
$q(u), Q(s)$	nonnegative remainder at $u=1$ and its transform	(7.1)
$\nu_\sigma, \bar{v}_\sigma, V(\sigma)$	probability measure, its mean and its variance	(8.3), (8.2)
δ, R_δ	half-opening defect of a sector and its radius	Theorem 9.1
$v_0, \varepsilon(v)$	cut point $\frac{1}{2}$ and local remainder	(9.3)
$\mathsf{M}, \mathsf{N}, \mathsf{L}$	truncation integers: harmonics, periods, nodes	Definition 10.1
$\mathcal{K}_\mathsf{M}, H_{\mathsf{M},\mathsf{N}}, H_{\mathsf{M},\mathsf{N},\mathsf{L}}$	truncated kernel and truncated transforms	(10.1)–(10.2)
x_b	quadrature nodes $(b+\frac{1}{2})/\mathsf{L}$	(10.2)
B_0, B_1, B_2	bounds for $\mathcal{K}_\mathsf{M}$, $\mathcal{K}'_\mathsf{M}$, $\mathcal{K}''_\mathsf{M}$	(10.3)
$Z(\theta)$	$1+1/(\theta-1)$, a bound for $\sum_n n^{-\theta}$	(10.3)
E_0, E_1	error bounds for H and H'	(10.5), (10.6)
χ_n	integrand on the n-th period	proof of Theorem 10.2
s_0, ς	centre and radius of a disc	Proposition 10.4
η_0, η_1	computed values of H and H' at s_0	Proposition 10.4
$J_2(d)$	Jordan totient of order 2	(11.3)
c_k, N	real coefficients and their number	Corollary 11.2
K	number of factors	Corollary 11.3
$\mathrm{har}(n)$	harmonic number $\sum_{b\leq n} 1/b$	proof of Corollary 11.3
g, j', k'	$g=(j,k)$, $j=gj'$, $k=gk'$	Section 12
$I_{jk}(\sigma)$	weighted Gram entry	(12.1)
$T_{jk}(r), U_{jk}(r), D_{jk}(r)$	pair sums	(12.2), (12.3)
$m_*, n_*, \alpha, \beta, m_0, n_0, l_0$	least solution and auxiliary quantities of Lemma 12.2	Lemma 12.2
F, F_0, F_1, F_2, F_3	$h(x/j)h(x/k)$, $F-T_{jk}(0)$, $h(x/j)$, $h(x/k)$, $F_2 e^{2\pi i \mathfrak{q} x/(jk)}$	Lemma 12.3
$\kappa_\sigma(y)$	kernel of the weighted Gram matrix, $y>0$	(12.10)
$a, \omega, w(x)$	$j\vee k$, $2\pi/(jk)$, $x^{-2\sigma-1}$	Section 12
Y, Y_0, Y_1	primitives of F_0	proofs of Proposition 12.5 and Theorem 12.6
X	upper limit of integration	(12.14)
$A(s)$	auxiliary function of the figures	(14.1)
Σ_{2k}	$\sum_{m\geq 1} m^{-2k}$ for the integers $k\geq 1$	Proposition 14.1
$\widetilde{H}(s)$	transform of $\mathcal{K}$ on $(0,1)$	(14.4)
z_b^G, n_G	computed zeros of G and their number	Section 14

Symbol	Meaning	Defined in
z_b^A, n_A	computed real zeros of A and their number	Section 14
$\mathcal{E}_G(x), \mathcal{E}_A(x)$	exponential sums over these zeros	(14.8)
$\zeta(s)$	Riemann zeta function, used only in Sections 14, 17 and 18	Section 14
$\ell(u)$	$\log(\pi u/\lvert\sin(\pi u)\rvert)$, $u > 0$	(15.1)
$q_0(u), Q_0(s)$	remainder on $(0,1)$ and its transform	(15.4), (15.5)
$\mathcal{Q}(t), \mathfrak{r}_k$	upper bound for $\mathrm{Re}(sA(s))$, real part of $s/(2k-s)$	Theorem 15.4
$\mathfrak{b}$	$\log(2\pi) - \frac{3}{2}$	Section 16
$\mathcal{B}(u), \Omega(s)$	positive convex kernel of $G - A$ and its Mellin transform	(16.1)
$\upsilon_\sigma(v), \upsilon_\sigma^{\pm}(v)$	weights of Ω in the variable $v = \log u$	(16.7)
$\mathcal{M}_j(\sigma)$	moments of $\mathcal{B}$, $j = 0, 1, 2$	Theorem 16.3
$\mathcal{K}^\sharp, H^\sharp, \widetilde{H}^\sharp, G^\sharp, A^\sharp, \mathfrak{c}_n$	kernel with two harmonics and its counterparts of $H, \widetilde{H}, G, A$; Taylor coefficients	(16.12)
$\mathcal{C}(s)$	$\int_0^\infty (1-\cos y) y^{-s-3} dy$	(16.13)
$\Gamma(s)$	Euler's gamma function, used only in Section 17	Section 17
$N(T), N_0(T)$	number of zeros of ζ up to height T, main term plus $\frac{7}{8}$	(17.6)
τ_k, W	solutions of $N_0(\tau_k) = k - \frac{1}{2}$, inverse of $w \mapsto we^w$	(17.7)

3. The kernel h and the transform G

Reviewed by Claude (review 2)

Lemma 3.1 (The kernel h)**.** *The function h of (1.1) has period 1, is even, and satisfies $\int_0^1 \lvert h(u)\rvert^n \, du < \infty$ for every integer $n \geq 1$. Its Fourier coefficients with respect to the period 1 are*

$$\widehat{h}_1(0) = 0, \qquad \widehat{h}_1(m) = -\frac{1}{2|m|} \quad (m \in \mathbb{Z},\ m \neq 0), \tag{3.1}$$

so that $h(u) = -\sum_{m\geq 1} \cos(2\pi m u)/m$ in $L^2(0,1)$. Moreover

$$\int_0^1 h(u)^2 \, du = \frac{\pi^2}{12}. \tag{3.2}$$

Proof. Periodicity and evenness follow from $\lvert\sin(\pi(u+1))\rvert = \lvert\sin(\pi u)\rvert = \lvert\sin(-\pi u)\rvert$. For $0 < u \leq \frac{1}{2}$ we have $2u \leq \sin(\pi u) \leq 1$, hence $\log(4u) \leq h(u) \leq \log 2$ and $\lvert h(u)\rvert \leq \log 2 + \lvert\log(4u)\rvert$. Since $h(1-u) = h(u)$, the same bound holds with $1-u$ in place of u for $\frac{1}{2} \leq u < 1$. As $\int_0^1 \lvert\log u\rvert^n \, du < \infty$ for every $n \geq 1$, the integrability follows.

Let $0 < \varrho < 1$. The series $-\sum_{m\geq 1} \varrho^m e^{2\pi i m u}/m$ converges absolutely to the principal logarithm of $1 - \varrho e^{2\pi i u}$. Taking real parts gives

$$\log\lvert 1 - \varrho e^{2\pi i u}\rvert = -\sum_{m\geq 1} \frac{\varrho^m}{m} \cos(2\pi m u) \qquad (u \in \mathbb{R}). \tag{3.3}$$

Let ϕ_ϱ denote the right side of (3.3) and let $\phi = -\sum_{m\geq 1} \cos(2\pi m \cdot)/m$, a series that converges in $L^2(0,1)$. By (2.2), $\int_0^1 (\phi_\varrho - \phi)^2 = \frac{1}{2}\sum_{m\geq 1} (1-\varrho^m)^2/m^2$, which tends to 0 as $\varrho \to 1$ by dominated convergence. Hence there is a sequence of values of ϱ tending to 1 along which $\phi_\varrho \to \phi$ almost everywhere. On the other hand, for $u \notin \mathbb{Z}$ the left side of (3.3) tends to

$\log|1-e^{2\pi iu}| = \log|2\sin(\pi u)| = h(u)$ as $\varrho \to 1$. Therefore $h = \phi$ almost everywhere, which is (3.1). Finally (2.2) with $\Phi_1 = \Phi_2 = h$ and $L = 1$ gives $\int_0^1 h^2 = \sum_{m\neq 0} 1/(4m^2) = \frac{1}{2}\sum_{m\geq 1} 1/m^2 = \pi^2/12$, by Euler's evaluation $\sum_{m\geq 1} 1/m^2 = \pi^2/6$. □

For $\operatorname{Re} s = \sigma > 0$ the integral (1.2) converges absolutely, because $\int_n^{n+1}|h| = \int_0^1 |h|$ for every integer $n \geq 1$ and $\sum_{n\geq 1} n^{-\sigma-1} < \infty$. Hence G is holomorphic in $\operatorname{Re} s > 0$ and $G(\overline{s}) = \overline{G(s)}$.

Reviewed by Claude (review 1)

Proposition 3.2 (Local factor; classical)**.** *Let $\sigma > 0$ and let $\xi(k)$, $k \geq 1$, be complex numbers with $\sum_{k\geq 1}|\xi(k)|k^{-\sigma} < \infty$. For real $x \geq 1$ that is not an integer multiple of any integer $k \leq x$ put*

$$S_\xi(x) = \sum_{1\leq k\leq x} \xi(k)\, h(x/k). \tag{3.4}$$

Then for $\operatorname{Re} s = \sigma$,

$$\int_1^\infty S_\xi(x)\, x^{-s-1}\, dx = G(s) \sum_{k\geq 1} \frac{\xi(k)}{k^s}, \tag{3.5}$$

and the integral converges absolutely.

Proof. The excluded values of x form a countable set. The substitution $x = ku$ gives

$$\int_k^\infty |h(x/k)|\, x^{-\sigma-1}\, dx = k^{-\sigma} \int_1^\infty |h(u)|\, u^{-\sigma-1}\, du. \tag{3.6}$$

Summing over k with the weights $|\xi(k)|$ gives a finite total, and Fubini's theorem allows the interchange of the sum over k with the integral over $x \geq k$. The same substitution without absolute values gives (3.5). □

4. The positive periodic kernel

For real x put

$$\mathcal{K}(x) = \sum_{m\geq 1} \frac{1-\cos(2\pi mx)}{m^3}, \qquad \Sigma_3 = \sum_{m\geq 1} \frac{1}{m^3}. \tag{4.1}$$

Reviewed by Claude (review 1)

Proposition 4.1 (Positivity and boundary data)**.** *The series* (4.1) *converges uniformly. The function $\mathcal{K}$ is real, even, has period* 1 *and is continuously differentiable, with*

$$0 \leq \mathcal{K}(x) \leq 2\Sigma_3 < \frac{5}{2}, \qquad \mathcal{K}(x) > 0 \quad (x \notin \mathbb{Z}), \qquad \mathcal{K}(n) = \mathcal{K}'(n) = 0 \quad (n \in \mathbb{Z}), \tag{4.2}$$

and $\int_0^1 \mathcal{K} = \Sigma_3$. Moreover $\mathcal{K}'$ is absolutely continuous and

$$\mathcal{K}'(x) = -4\pi^2 \int_0^x h(u)\, du \qquad (x \in \mathbb{R}). \tag{4.3}$$

Proof. Each term of (4.1) lies in $[0, 2/m^3]$, so the series converges uniformly and $0 \leq \mathcal{K} \leq 2\Sigma_3$. Since $\sum_{m\geq 3} m^{-3} < \int_2^\infty x^{-3}dx = \frac{1}{8}$, we have $\Sigma_3 < 1 + \frac{1}{8} + \frac{1}{8}$ and $2\Sigma_3 < \frac{5}{2}$. For $x \notin \mathbb{Z}$ the term $m = 1$ is positive. The series of derivatives $2\pi\sum_{m\geq 1}\sin(2\pi mx)/m^2$ converges uniformly, so $\mathcal{K}$ is continuously differentiable and $\mathcal{K}'$ is that series. At integers every term of both series vanishes. Termwise integration gives $\int_0^1 \mathcal{K} = \Sigma_3$.

Let $\Phi(x) = -4\pi^2 \int_0^x h$. Since $\widehat{h}_1(0) = 0$, Φ has period 1, and it is absolutely continuous. For an integer $m \neq 0$, integration by parts over one period and (3.1) give $\widehat{\Phi}_1(m) = -4\pi^2\widehat{h}_1(m)/(2\pi im) = \pi/(im|m|)$. The Fourier coefficient of $\mathcal{K}'$ at $m \neq 0$ is $2\pi \cdot \frac{1}{2i} \cdot \frac{m}{|m|} \cdot \frac{1}{m^2} = \pi/(im|m|)$ as well. Hence $\Phi - \mathcal{K}'$ is continuous with period 1 and its Fourier coefficients at all $m \neq 0$ vanish, so it is constant by the remark after (2.2). Since $\Phi(0) = 0 = \mathcal{K}'(0)$, we get $\Phi = \mathcal{K}'$. □

Reviewed by Claude (review 1)

Lemma 4.2 (Local form of the kernel)**.** *For $0 < x \le \frac{1}{2}$,*

$$\Big|\mathcal{K}(x) - 2\pi^2 x^2\Big(\log\frac{1}{x} + \frac{3}{2} - \log(2\pi)\Big)\Big| \le \frac{\pi^4 x^4}{9}. \tag{4.4}$$

Proof. By (4.3) and $\mathcal{K}(0) = 0$, $\mathcal{K}(x) = -4\pi^2 \int_0^x (x-\tau)h(\tau)\,d\tau$. For $0 < \tau \le \frac{1}{2}$ write $h(\tau) = \log(2\pi\tau) + \log(\sin(\pi\tau)/(\pi\tau))$. For $0 < \theta \le \frac{\pi}{2}$ we have $1 - \theta^2/6 \le \sin\theta/\theta \le 1$ and $\theta^2/6 \le \frac{1}{2}$, and $\log(1-\lambda) \ge -2\lambda$ for $0 \le \lambda \le \frac{1}{2}$. Hence $|\log(\sin(\pi\tau)/(\pi\tau))| \le \pi^2\tau^2/3$. A direct integration gives $\int_0^x (x-\tau)\log(2\pi\tau)\,d\tau = \frac{x^2}{2}\log(2\pi x) - \frac{3x^2}{4}$, and $\int_0^x (x-\tau)\pi^2\tau^2/3\,d\tau = \pi^2 x^4/36$. Multiplying by $4\pi^2$ gives (4.4). □

5. Factorization and analytic continuation

Reviewed by Claude (review 1)

Theorem 5.1 (Factorization through a positive kernel)**.** *The integral*

$$H(s) = \int_1^\infty \mathcal{K}(u)\,u^{-s-3}\,du = \int_0^1 \mathcal{K}(x)\sum_{n\ge 1}(n+x)^{-s-3}\,dx \tag{5.1}$$

converges absolutely for $\operatorname{Re} s > -2$ *and defines a holomorphic function there. For* $\operatorname{Re} s > 0$,

$$G(s) = -\frac{(s+1)(s+2)}{4\pi^2}\,H(s). \tag{5.2}$$

Proof. The convergence follows from $0 \le \mathcal{K} \le \frac{5}{2}$. The second form of H follows by splitting $[1,\infty)$ into the intervals $[n, n+1]$ and using the period 1 of $\mathcal{K}$. Let $\operatorname{Re} s > 0$ and let $X \ge 2$ be an integer. By (4.3), $h = -\mathcal{K}''/(4\pi^2)$ almost everywhere, with $\mathcal{K}'$ absolutely continuous. Two integrations by parts give

$$\int_1^X h(u)\,u^{-s-1}\,du = -\frac{s+1}{4\pi^2}\int_1^X \mathcal{K}'(u)\,u^{-s-2}\,du = -\frac{(s+1)(s+2)}{4\pi^2}\int_1^X \mathcal{K}(u)\,u^{-s-3}\,du. \tag{5.3}$$

There are no boundary terms because $\mathcal{K}'(1) = \mathcal{K}'(X) = 0$ and $\mathcal{K}(1) = \mathcal{K}(X) = 0$ by (4.2). Letting $X \to \infty$ gives (5.2). □

For a complex parameter ϖ with $\operatorname{Re}\varpi > \frac{1}{2}$ and a bounded function f on $(0,1]$ put

$$(\mathcal{L}_\varpi f)(x) = \sum_{n\ge 1}\frac{1}{(n+x)^{2\varpi}}\,f\Big(\frac{1}{n+x}\Big) \qquad (0 \le x \le 1). \tag{5.4}$$

For $\varpi = 1$ this is the transfer operator of the Gauss map $x \mapsto \{1/x\}$ acting on densities. By (5.1), for $\operatorname{Re} s > -2$,

$$H(s) = \int_0^1 \mathcal{K}(x)\,(\mathcal{L}_{(s+3)/2}\mathbf{1})(x)\,dx, \tag{5.5}$$

where $\mathbf{1}$ is the constant function 1. Thus the zeros of G other than -1 and -2 in $\operatorname{Re} s > -2$ are the zeros of a matrix element of the family $\mathcal{L}_\varpi$.

Reviewed by Claude (review 1)

Theorem 5.2 (Entire continuation and special values)**.** *The function H extends to a meromorphic function on $\mathbb{C}$ whose only pole is a simple pole at $s = -2$ with residue Σ_3. The function G extends to an entire function, the identity* (5.2) *holds on $\mathbb{C}$, and with $\Sigma_5 = \sum_{m\ge1} m^{-5}$,*

$$G(-1) = 0, \qquad G(-2) = \frac{\Sigma_3}{4\pi^2}, \qquad G(-3) = \frac{\Sigma_3}{2\pi^2}, \qquad G(-4) = \frac{3\Sigma_3}{4\pi^2} - \frac{3\Sigma_5}{8\pi^4}. \tag{5.6}$$

Proof. For $\operatorname{Re} s > -2$ and an integer $m \ge 1$ let $C_m(s) = \int_1^\infty \cos(2\pi m u)u^{-s-3}du$. Inserting (4.1) in (5.1), with absolute convergence,

$$H(s) = \frac{\Sigma_3}{s+2} - \sum_{m\ge1}\frac{C_m(s)}{m^3} \qquad (\operatorname{Re} s > -2). \tag{5.7}$$

Two integrations by parts, in which $\sin(2\pi m)=0$ and $\cos(2\pi m)=1$, give for $\operatorname{Re} s > -2$

$$C_m(s) = \frac{s+3}{2\pi m}\int_1^\infty \sin(2\pi m u)\, u^{-s-4}\, du = \frac{s+3}{(2\pi m)^2} - \frac{(s+3)(s+4)}{(2\pi m)^2}\, C_m(s+2). \tag{5.8}$$

The first equality gives $|C_m(s)| \le |s+3|/(2\pi m(\sigma+3))$ for $\sigma > -2$. The right side of (5.8) is holomorphic for $\operatorname{Re} s > -4$, so it extends C_m there, and by induction C_m extends to an entire function that satisfies (5.8) on $\mathbb{C}$. By the same induction, on every compact subset of $\mathbb{C}$ there is a constant, independent of m, that bounds $m\,|C_m(s)|$. Hence $\sum_m C_m(s)/m^3$ converges uniformly on compact sets and is entire. By (5.7), H is meromorphic on $\mathbb{C}$ with a single simple pole at -2 of residue Σ_3. The function $-(s+1)(s+2)H(s)/(4\pi^2)$ is therefore entire, and it equals G for $\operatorname{Re} s > 0$ by (5.2).

Since H is finite at -1, $G(-1)=0$. At $s=-2$ the factor $s+2$ cancels the pole and $G(-2) = -(-1)\Sigma_3/(4\pi^2)$. By (5.8), $C_m(-3)=0$ and $C_m(-4) = -1/(2\pi m)^2$. Hence $H(-3) = -\Sigma_3$ and $H(-4) = -\Sigma_3/2 + \Sigma_5/(4\pi^2)$ by (5.7), and (5.2) gives $G(-3) = -2H(-3)/(4\pi^2)$ and $G(-4) = -6H(-4)/(4\pi^2)$, which are the last two values in (5.6). □

6. The real axis

Reviewed by Claude (review 1)

Theorem 6.1 (Sign on the real axis)**.** *For real s,*

$$G(s) > 0 \quad (-2 < s < -1), \qquad G(-1) = 0, \qquad G(s) < 0 \quad (s > -1). \tag{6.1}$$

The zero at -1 is simple, and

$$G'(-1) = -\frac{1}{4\pi^2}\int_1^\infty \mathcal{K}(u)\, u^{-2}\, du < 0. \tag{6.2}$$

Proof. For real $s > -2$ the integrand of (5.1) is nonnegative and positive off the integers, so $H(s) > 0$. By (5.2) the sign of $G(s)$ is the sign of $-(s+1)(s+2)$. Differentiating (5.2) at -1 gives $G'(-1) = -H(-1)/(4\pi^2)$. □

7. The singularity at $u=1$ and a completely monotone remainder

For real $u > 1$, $u \notin \mathbb{Z}$, and for $\operatorname{Re} s > 0$ put

$$q(u) = \log\frac{\pi(u-1)}{|\sin(\pi u)|}, \qquad Q(s) = \int_1^\infty q(u)\, u^{-s-1}\, du. \tag{7.1}$$

Reviewed by Claude (review 1)

Proposition 7.1 (Remainder of constant sign)**.** *We have $q(u) \ge 0$, the integral $Q(s)$ converges absolutely for $\operatorname{Re} s > 0$, and*

$$G(s) = \frac{\log(2\pi) - \gamma - \psi(s)}{s} - Q(s) \qquad (\operatorname{Re} s > 0). \tag{7.2}$$

For real $\sigma > 0$ and real t,

$$(-1)^n Q^{(n)}(\sigma) > 0 \quad (n \ge 0), \qquad |Q(\sigma+it)| \le Q(\sigma). \tag{7.3}$$

Proof. Since $|\sin(\pi u)| = |\sin(\pi(u-1))| \le \pi(u-1)$, we get $q \ge 0$, and $q > 0$ for $u > 2$. By (1.1), $h(u) = \log(2\pi(u-1)) - q(u)$. The function $\log(2\pi(u-1))u^{-\sigma-1}$ is absolutely integrable on $[1,\infty)$, and so is $h(u)u^{-\sigma-1}$, hence so is $q(u)u^{-\sigma-1}$. Therefore

$$G(s) = \frac{\log(2\pi)}{s} + \int_1^\infty \log(u-1)\, u^{-s-1}\, du - Q(s). \tag{7.4}$$

The substitution $x = 1/u$ turns the middle integral into $\int_0^1 x^{s-1}(\log(1-x) - \log x)\, dx$. Termwise integration of $\log(1-x) = -\sum_{n\ge1} x^n/n$, which is justified by absolute convergence, gives $\int_0^1 x^{s-1}\log(1-x)\, dx = -\sum_{n\ge1} 1/(n(n+s)) = -\frac{1}{s}\sum_{n\ge1}(1/n - 1/(n+s))$. By (2.3) this is $-(\psi(s+1)+\gamma)/s = -(\psi(s)+\gamma)/s - 1/s^2$. Also $\int_0^1 x^{s-1}\log x\, dx = -1/s^2$. Hence the middle integral in (7.4) equals

$-(\psi(s)+\gamma)/s$, which proves (7.2). Differentiation under the integral sign gives $(-1)^n Q^{(n)}(\sigma) = \int_1^\infty q(u)(\log u)^n u^{-\sigma-1}du > 0$, and $|Q(\sigma+it)| \le \int_1^\infty q(u)u^{-\sigma-1}du = Q(\sigma)$. □

8. Moments, variance and a zero-free region

Reviewed by Claude (review 1)

Theorem 8.1 (Complete monotonicity). *For real $\sigma > -2$ and every integer $n \ge 0$,*

$$(-1)^n H^{(n)}(\sigma) = \int_1^\infty \mathcal{K}(u)\,(\log u)^n\, u^{-\sigma-3}\,du > 0, \tag{8.1}$$

and

$$V(\sigma) = \frac{H''(\sigma)}{H(\sigma)} - \Big(\frac{H'(\sigma)}{H(\sigma)}\Big)^2 > 0. \tag{8.2}$$

Proof. Differentiation under the integral sign in (5.1) is allowed on compact subsets of $\sigma > -2$ and gives (8.1). The substitution $u = e^v$ shows that

$$d\nu_\sigma(v) = \frac{\mathcal{K}(e^v)\,e^{-(\sigma+2)v}}{H(\sigma)}\,dv \qquad (v > 0) \tag{8.3}$$

is a probability measure on $(0,\infty)$, with mean $\bar v_\sigma = \int v\,d\nu_\sigma = -H'(\sigma)/H(\sigma)$ and second moment $H''(\sigma)/H(\sigma)$. Hence $V(\sigma)$ is the variance of ν_σ. It is positive because ν_σ has a density and is not a point mass. □

Reviewed by Claude (review 1)

Theorem 8.2 (Zero-free region from the variance). *Let $\sigma > -2$ and t be real with $\sigma + it \ne -1$. If $G(\sigma+it) = 0$, then*

$$t^2\,V(\sigma) \ge 2. \tag{8.4}$$

Consequently $G(\sigma+it) \ne 0$ whenever $\sigma > -2$, $\sigma + it \ne -1$ and $|t| < \sqrt{2/V(\sigma)}$.

Proof. By Theorem 5.2, $G(\sigma+it) = 0$ with $\sigma > -2$ and $\sigma + it \ne -1$ forces $H(\sigma+it) = 0$. By (5.1) and (8.3), $H(\sigma+it)/H(\sigma) = \int e^{-itv}\,d\nu_\sigma(v)$. If this vanishes, then so does the real part of its product with $e^{it\bar v_\sigma}$, that is, $\int \cos(t(v-\bar v_\sigma))\,d\nu_\sigma(v) = 0$. Since $\cos\theta \ge 1 - \theta^2/2$ for real θ, we get $0 \ge 1 - \frac{t^2}{2}\int (v-\bar v_\sigma)^2 d\nu_\sigma = 1 - \frac{t^2}{2}V(\sigma)$. □

The moments in (8.2) can be computed from (8.1). The computation described in Section 10 gives $H(\frac12) = 0.4179934$, $-H'(\frac12) = 0.2025230$, $H''(\frac12) = 0.1578961$ and $V(\frac12) = 0.1429955$, so that $\sqrt{2/V(\frac12)} = 3.7398$.

9. Expansion in right sectors

Reviewed by Claude (review 1)

Theorem 9.1 (Expansion in sectors). *Let $0 < \delta < \frac{\pi}{2}$. For s in the sector $|\arg s| \le \frac{\pi}{2} - \delta$, as $|s| \to \infty$,*

$$G(s) = \frac{\log(2\pi) - \gamma - \log s}{s} + \frac{1}{2s^2} + \frac{1/12 - \pi^2/3}{s^3} + O_\delta(|s|^{-4}), \tag{9.1}$$

with the principal branch of the logarithm. Consequently there is $R_\delta > 0$ such that $G(s) \ne 0$ for $|s| > R_\delta$ and $|\arg s| \le \frac{\pi}{2} - \delta$.

Proof. The substitution $u = e^v$ in (1.2) gives $G(s) = \int_0^\infty h(e^v)e^{-sv}dv$ for $\operatorname{Re} s > 0$. Put $v_0 = \frac12$. For $0 < v \le v_0$ we have $0 < \pi(e^v - 1) < \pi$ and $|\sin(\pi e^v)| = \sin(\pi(e^v-1))$, so

$$h(e^v) - \log(2\pi v) = \log\frac{e^v - 1}{v} + \log\frac{\sin(\pi(e^v-1))}{\pi(e^v-1)}. \tag{9.2}$$

The right side is real analytic on $[0, v_0]$. From $(e^v-1)/v = 1 + v/2 + v^2/6 + O(v^3)$ and $\log(\sin\theta/\theta) = -\theta^2/6 + O(\theta^4)$ its Taylor polynomial of degree 2 at 0 is $v/2 + (1/24 - \pi^2/6)v^2$. Hence there is a function ε and a constant B with

$$h(e^v) = \log(2\pi v) + \tfrac12 v + \Big(\frac{1}{24} - \frac{\pi^2}{6}\Big)v^2 + \varepsilon(v), \qquad |\varepsilon(v)| \le Bv^3 \quad (0 < v \le v_0). \tag{9.3}$$

Let $\sigma = \operatorname{Re} s \geq 1$. For $v \geq v_0$ we have $\sigma v \geq \sigma v_0/2 + v/2$. Therefore $\int_{v_0}^{\infty} |h(e^v)| e^{-\sigma v} dv \leq e^{-\sigma v_0/2} \int_1^{\infty} |h(u)| u^{-3/2} du$, and the integral over $v \geq v_0$ of $e^{-\sigma v}$ times $|\log(2\pi v)| + v/2 + |1/24 - \pi^2/6| v^2$ is at most a constant times $e^{-\sigma v_0/2}$. Also $|\int_0^{v_0} \varepsilon(v) e^{-sv} dv| \leq B \int_0^{\infty} v^3 e^{-\sigma v} dv = 6B/\sigma^4$. Hence, with an absolute implied constant,

$$G(s) = \int_0^{\infty} \Big(\log(2\pi v) + \tfrac{1}{2} v + \Big(\frac{1}{24} - \frac{\pi^2}{6} \Big) v^2 \Big) e^{-sv} \, dv + O\big(\sigma^{-4} + e^{-\sigma/4}\big). \tag{9.4}$$

For $\operatorname{Re} s > 0$ we have $\int_0^{\infty} v^n e^{-sv} dv = n!/s^{n+1}$ and $\int_0^{\infty} e^{-sv} \log v \, dv = -(\gamma + \log s)/s$. The second formula is the classical evaluation $\int_0^{\infty} e^{-v} \log v \, dv = -\gamma$ of Euler after the change of variable $v \mapsto v/s$ for real $s > 0$, and it extends to $\operatorname{Re} s > 0$ because both sides are holomorphic there. In the sector, $\sigma \geq |s| \sin \delta$, so the error term in (9.4) is $O_\delta(|s|^{-4})$, and (9.1) follows.

In (9.1) the modulus of the first term is at least $(\log|s| - |\log(2\pi) - \gamma|)/|s|$, because the real part of $\log s$ is $\log|s|$. The other terms are $O_\delta(|s|^{-2})$. Hence $G(s) \neq 0$ for $|s|$ large in the sector. □

Reviewed by Claude (review 1)

Corollary 9.2 (Angular restriction)**.** *If $\sigma_b + i t_b$, $b \geq 1$, are zeros of G with $\sigma_b \to +\infty$, then $|t_b|/\sigma_b \to +\infty$.*

Proof. If $|t_b|/\sigma_b$ stayed bounded along a subsequence, these zeros would lie in a fixed sector $|\arg s| \leq \frac{\pi}{2} - \delta$ with moduli tending to infinity. This contradicts Theorem 9.1. □

10. Truncation, quadrature and a Rouché test

Reviewed by Claude (review 1)

Definition 10.1 (Truncations)**.** Let $\mathsf{M}, \mathsf{N}, \mathsf{L} \geq 1$ be integers and $x_b = (b + \frac{1}{2})/\mathsf{L}$ for $0 \leq b \leq \mathsf{L} - 1$. Put

$$\mathcal{K}_{\mathsf{M}}(x) = \sum_{m=1}^{\mathsf{M}} \frac{1 - \cos(2\pi m x)}{m^3}, \qquad H_{\mathsf{M},\mathsf{N}}(s) = \sum_{n=1}^{\mathsf{N}} \int_0^1 \mathcal{K}_{\mathsf{M}}(x) \, (n + x)^{-s-3} \, dx, \tag{10.1}$$

$$H_{\mathsf{M},\mathsf{N},\mathsf{L}}(s) = \frac{1}{\mathsf{L}} \sum_{n=1}^{\mathsf{N}} \sum_{b=0}^{\mathsf{L}-1} \mathcal{K}_{\mathsf{M}}(x_b) \, (n + x_b)^{-s-3}. \tag{10.2}$$

Put also

$$B_0 = \frac{5}{2}, \qquad B_1 = \frac{\pi^3}{3}, \qquad B_2 = 4\pi^2(1 + \log \mathsf{M}), \qquad Z(\theta) = 1 + \frac{1}{\theta - 1} \quad (\theta > 1). \tag{10.3}$$

For $\theta > 1$ we have $\sum_{n \geq 1} n^{-\theta} \leq 1 + \int_1^{\infty} x^{-\theta} dx = Z(\theta)$. The functions $H_{\mathsf{M},\mathsf{N}}$ and $H_{\mathsf{M},\mathsf{N},\mathsf{L}}$ are entire. By (10.1), $0 \leq \mathcal{K}_{\mathsf{M}} \leq B_0$, $|\mathcal{K}'_{\mathsf{M}}| \leq 2\pi \sum_{m \geq 1} m^{-2} = B_1$ and $|\mathcal{K}''_{\mathsf{M}}| \leq 4\pi^2 \sum_{m \leq \mathsf{M}} 1/m \leq B_2$. We use the midpoint rule in the following form. If χ is a twice continuously differentiable complex function on $[0, 1]$, then

$$\Big| \int_0^1 \chi(x) \, dx - \frac{1}{\mathsf{L}} \sum_{b=0}^{\mathsf{L}-1} \chi(x_b) \Big| \leq \frac{\sup |\chi''|}{24 \, \mathsf{L}^2}. \tag{10.4}$$

Indeed, on the subinterval of length $1/\mathsf{L}$ centred at x_b, Taylor's formula gives $|\chi(x) - \chi(x_b) - \chi'(x_b)(x - x_b)| \leq \frac{1}{2} \sup |\chi''| (x - x_b)^2$, the linear term integrates to 0, and $\int (x - x_b)^2 dx = 1/(12\mathsf{L}^3)$ over that subinterval.

Reviewed by Claude (review 1)

Theorem 10.2 (Explicit error bounds)**.** *Let $\sigma = \operatorname{Re} s > -1$. Then $|H(s) - H_{\mathsf{M},\mathsf{N},\mathsf{L}}(s)| \leq E_0$ and $|H'(s) - H'_{\mathsf{M},\mathsf{N},\mathsf{L}}(s)| \leq E_1$, where*

$$\begin{aligned} E_0 = {} & \frac{1}{(\sigma + 2)\mathsf{M}^2} + \frac{B_0}{(\sigma + 2)(\mathsf{N} + 1)^{\sigma + 2}} \\ & + \frac{B_2 Z(\sigma + 3) + 2|s + 3| B_1 Z(\sigma + 4) + |(s + 3)(s + 4)| B_0 Z(\sigma + 5)}{24 \, \mathsf{L}^2} \end{aligned} \tag{10.5}$$

and

$$E_1 = \frac{1}{(\sigma+2)^2\mathsf{M}^2} + \frac{B_0}{(\mathsf{N}+1)^{\sigma+2}}\Big(\frac{\log(\mathsf{N}+1)}{\sigma+2} + \frac{1}{(\sigma+2)^2}\Big) + \frac{B_2 Z(\sigma+2) + 2B_1\big(Z(\sigma+4) + |s+3|Z(\sigma+3)\big)}{24\,\mathsf{L}^2} + \frac{B_0\big(|(s+3)(s+4)|Z(\sigma+4) + |2s+7|Z(\sigma+5)\big)}{24\,\mathsf{L}^2}. \tag{10.6}$$

Proof. By (4.1) and (10.1), $0 \le \mathcal{K} - \mathcal{K}_\mathsf{M} \le 2\sum_{m>\mathsf{M}} m^{-3} \le 2\int_\mathsf{M}^\infty x^{-3}dx = \mathsf{M}^{-2}$. The difference $H - H_{\mathsf{M},\mathsf{N}}$ is the integral of $(\mathcal{K}-\mathcal{K}_\mathsf{M})(u)u^{-s-3}$ over $1 \le u \le \mathsf{N}+1$ plus the integral of $\mathcal{K}(u)u^{-s-3}$ over $u \ge \mathsf{N}+1$. With $\int_1^\infty u^{-\sigma-3}du = 1/(\sigma+2)$, $\int_{\mathsf{N}+1}^\infty u^{-\sigma-3}du = (\mathsf{N}+1)^{-\sigma-2}/(\sigma+2)$ and $\mathcal{K} \le B_0$, this gives the first two terms of E_0. For the third, apply (10.4) to $\chi_n(x) = \mathcal{K}_\mathsf{M}(x)(n+x)^{-s-3}$ for $1 \le n \le \mathsf{N}$. Its second derivative is $\mathcal{K}''_\mathsf{M}(n+x)^{-s-3} - 2(s+3)\mathcal{K}'_\mathsf{M}(n+x)^{-s-4} + (s+3)(s+4)\mathcal{K}_\mathsf{M}(n+x)^{-s-5}$, and $|(n+x)^{-s-\theta}| \le n^{-\sigma-\theta}$ for $0 \le x \le 1$ and $\sigma+\theta > 0$. Summing over n with $\sum_n n^{-\theta} \le Z(\theta)$ gives the third term of E_0.

For the derivative, the integrand is $-\mathcal{K}(u)(\log u)u^{-s-3}$. We use $\int_1^\infty (\log u)u^{-\sigma-3}du = (\sigma+2)^{-2}$ and, for $X \ge 1$, $\int_X^\infty (\log u)u^{-\sigma-3}du = X^{-\sigma-2}(\log X/(\sigma+2) + (\sigma+2)^{-2})$, with $X = \mathsf{N}+1$. This gives the first two terms of E_1. For the third, apply (10.4) to $\chi_n(x) = -\mathcal{K}_\mathsf{M}(x)\log(n+x)\,(n+x)^{-s-3}$. For $u \ge 1$,

$$\begin{aligned}\frac{d}{du}\big[(\log u)u^{-s-3}\big] &= u^{-s-4}\big(1 - (s+3)\log u\big),\\ \frac{d^2}{du^2}\big[(\log u)u^{-s-3}\big] &= u^{-s-5}\big((s+3)(s+4)\log u - (2s+7)\big).\end{aligned} \tag{10.7}$$

With $0 \le \log u \le u$, the moduli of $(\log u)u^{-s-3}$ and of the two derivatives in (10.7) are at most $u^{-\sigma-2}$, $u^{-\sigma-4} + |s+3|u^{-\sigma-3}$ and $|(s+3)(s+4)|u^{-\sigma-4} + |2s+7|u^{-\sigma-5}$. These bounds decrease in u because $\sigma > -1$, so on the n-th period they are at most their values at $u = n$. The product rule and $\sum_n n^{-\theta} \le Z(\theta)$ give the third term of E_1. □

Lemma 10.3 (Second derivative)**.** *If* $\operatorname{Re} s \ge \sigma > -2$ *with* σ *real, then* $|H''(s)| \le 2B_0/(\sigma+2)^3$. Reviewed by Claude (review 1)

Proof. By (5.1) and $0 \le \mathcal{K} \le B_0$, $|H''(s)| \le B_0\int_1^\infty (\log u)^2 u^{-\sigma-3}du = 2B_0/(\sigma+2)^3$. □

Proposition 10.4 (Rouché test for a disc)**.** *Let* $s_0 \in \mathbb{C}$ *and* $\varsigma > 0$ *with* $\operatorname{Re} s_0 - \varsigma > -1$. *Let* $\eta_0 = H_{\mathsf{M},\mathsf{N},\mathsf{L}}(s_0)$ *and* $\eta_1 = H'_{\mathsf{M},\mathsf{N},\mathsf{L}}(s_0)$, *and let* E_0, E_1 *be the bounds of Theorem 10.2 at* $s = s_0$. *If* Reviewed by Claude (review 1)

$$|\eta_0| + E_0 + \varsigma E_1 + \frac{\varsigma^2 B_0}{(\operatorname{Re} s_0 - \varsigma + 2)^3} < \varsigma\,|\eta_1|, \tag{10.8}$$

then H *has exactly one zero in the disc* $|s - s_0| < \varsigma$, *and it is simple. It is also a simple zero of* G, *and* G *has no other zero in the disc.*

Proof. For $|s-s_0| = \varsigma$, Taylor's formula with integral remainder and Lemma 10.3 with $\sigma = \operatorname{Re} s_0 - \varsigma$ give $|H(s) - H(s_0) - H'(s_0)(s-s_0)| \le \frac{\varsigma^2}{2}\cdot 2B_0/(\operatorname{Re} s_0 - \varsigma + 2)^3$. By Theorem 10.2, $|H(s_0) - \eta_0| \le E_0$ and $|H'(s_0) - \eta_1| \le E_1$. Hence $|H(s) - \eta_1(s-s_0)|$ is at most the left side of (10.8), which is less than $\varsigma|\eta_1| = |\eta_1(s-s_0)|$. By Rouché's theorem H has in the disc as many zeros, counted with multiplicity, as the function $\eta_1(s-s_0)$, namely one. The disc lies in $\operatorname{Re} s > -1$, where the factor $(s+1)(s+2)$ of (5.2) does not vanish. □

Table 2 reports the test (10.8) at six points s_0, with $\mathsf{M} = 4000$, $\mathsf{N} = 200$, $\mathsf{L} = 65536$ and $\varsigma = 10^{-2}$. The six centres are the decimal numbers printed in the table, taken as exact rational numbers. The finite sums η_0 and η_1, the bounds E_0 and E_1 and the two sides of (10.8) were enclosed in ball arithmetic with the library Arb [1], at a precision of 128 bits. In this arithmetic every operation returns a midpoint and a radius such that the ball contains the exact result, so the rounding errors

are inside the radii. The radii of the real and imaginary parts of η_0 and η_1 are below 10^{-32}. At each of the six points the upper bound of the left side of (10.8) is smaller than the lower bound of the right side, and the left side is less than 0.47 times the right side. By Proposition 10.4, each disc $|s - s_0| < 10^{-2}$ contains exactly one zero of G, and this zero is simple. The entries of Table 2 are the enclosures rounded to the digits shown. The six centres have real parts between 0.2277 and 1.1273.

TABLE 2. The Rouché test (10.8) with $\mathsf{M} = 4000$, $\mathsf{N} = 200$, $\mathsf{L} = 65536$, $\varsigma = 10^{-2}$. The last two columns are the two sides of (10.8).

s_0	$\lvert\eta_0\rvert$	$\lvert\eta_1\rvert$	E_0	left side	right side
$1.127312874 + 15.001263432\,i$	$4.2 \cdot 10^{-9}$	0.018463	$8.6 \cdot 10^{-8}$	$8.3 \cdot 10^{-6}$	$1.8 \cdot 10^{-4}$
$0.533572290 + 21.763254707\,i$	$7.9 \cdot 10^{-8}$	0.011815	$1.5 \cdot 10^{-6}$	$1.7 \cdot 10^{-5}$	$1.2 \cdot 10^{-4}$
$0.894730810 + 25.624836921\,i$	$9.2 \cdot 10^{-9}$	0.006881	$2.4 \cdot 10^{-7}$	$1.1 \cdot 10^{-5}$	$6.9 \cdot 10^{-5}$
$0.227688021 + 30.975962723\,i$	$2.9 \cdot 10^{-7}$	0.006866	$8.4 \cdot 10^{-6}$	$3.2 \cdot 10^{-5}$	$6.9 \cdot 10^{-5}$
$0.950506780 + 33.599087545\,i$	$5.2 \cdot 10^{-9}$	0.003138	$2.0 \cdot 10^{-7}$	$1.0 \cdot 10^{-5}$	$3.1 \cdot 10^{-5}$
$0.430899874 + 38.054886210\,i$	$8.0 \cdot 10^{-8}$	0.004536	$2.7 \cdot 10^{-6}$	$2.1 \cdot 10^{-5}$	$4.5 \cdot 10^{-5}$

11. The mean over a period: a GCD matrix

Reviewed by Claude (review 2)

Theorem 11.1 (Gram matrix for the mean)**.** *Let $j, k \geq 1$ be integers. The function $x \mapsto h(x/j)h(x/k)$ is locally integrable on $\mathbb{R}$, has period $[j,k]$, and*

$$\operatorname{av}\big[h(x/j)\,h(x/k)\big] = \frac{\pi^2}{12} \cdot \frac{(j,k)^2}{jk}. \tag{11.1}$$

Proof. Put $L = [j,k]$, $g = (j,k)$, $j = gj'$ and $k = gk'$, so that $(j',k') = 1$ and $L = gj'k'$. The functions $F_1(x) = h(x/j)$ and $F_2(x) = h(x/k)$ have periods j and k, hence both have period L, and both are square integrable on $[0,L]$ by Lemma 3.1. Their product is therefore integrable on $[0,L]$ and has period L.

By (3.1), $F_1(x) = \sum_{m\neq 0} \widehat{h}_1(m) e^{2\pi i m x/j}$ in $L^2(0,L)$, and $e^{2\pi i m x/j} = e^{2\pi i (mL/j)x/L}$ with $mL/j = mk'$. Hence $(\widehat{F_1})_L(\mathfrak{q}) = \widehat{h}_1(m)$ if $\mathfrak{q} = mk'$ with $m \neq 0$, and $(\widehat{F_1})_L(\mathfrak{q}) = 0$ otherwise. In the same way $(\widehat{F_2})_L(\mathfrak{q}) = \widehat{h}_1(n)$ if $\mathfrak{q} = nj'$ with $n \neq 0$, and it is 0 otherwise. These coefficients are real. By (2.2),

$$\operatorname{av}[F_1 F_2] = \sum_{\substack{m,n\neq 0 \\ mk' = nj'}} \frac{1}{4|m|\,|n|}. \tag{11.2}$$

Since $(j',k') = 1$, the equation $mk' = nj'$ with $m, n \neq 0$ holds exactly when $m = j'l$ and $n = k'l$ for an integer $l \neq 0$. The sum in (11.2) is therefore $\sum_{l\neq 0} 1/(4j'k'l^2) = \pi^2/(12j'k')$, and $1/(j'k') = g^2/(jk)$. □

The Jordan totient of order 2 is the function defined for integers $d \geq 1$ by

$$J_2(d) = d^2 \prod_{p\mid d} \Big(1 - \frac{1}{p^2}\Big), \tag{11.3}$$

the product running over the primes p that divide d, so that $J_2(1) = 1$. It is multiplicative, and for a prime power p^k with $k \geq 1$ we have $\sum_{b=0}^{k} J_2(p^b) = 1 + \sum_{b=1}^{k}(p^{2b} - p^{2b-2}) = p^{2k}$. If $n = n_1 n_2$ with

$(n_1,n_2)=1$, every divisor of n is uniquely a product d_1d_2 with $d_1\mid n_1$ and $d_2\mid n_2$, so the function $n\mapsto\sum_{d\mid n}J_2(d)$ is multiplicative too. Hence

$$\sum_{d\mid n}J_2(d)=n^2\qquad(n\ge1).\tag{11.4}$$

Corollary 11.2 (Sum of squares)**.** *Let $N\ge1$ be an integer and let $c_1,\dots,c_N$ be real numbers. Then*

Reviewed by Claude (review 2)

$$\operatorname{av}\Big[\Big(\sum_{k=1}^{N}c_k\,h(x/k)\Big)^2\Big]=\frac{\pi^2}{12}\sum_{d=1}^{N}J_2(d)\Big(\sum_{\substack{1\le k\le N\\ d\mid k}}\frac{c_k}{k}\Big)^2.\tag{11.5}$$

Proof. All the functions $h(x/k)$ with $1\le k\le N$ have the common period $[1,2,\dots,N]$, so the mean of the square is the sum of $c_jc_k\operatorname{av}[h(x/j)h(x/k)]$ over $1\le j,k\le N$. By (11.1) and (11.4) with $n=(j,k)$,

$$\sum_{j,k=1}^{N}c_jc_k\frac{\pi^2}{12}\frac{(j,k)^2}{jk}=\frac{\pi^2}{12}\sum_{j,k=1}^{N}\frac{c_j}{j}\frac{c_k}{k}\sum_{\substack{d\mid j\\ d\mid k}}J_2(d)=\frac{\pi^2}{12}\sum_{d=1}^{N}J_2(d)\Big(\sum_{\substack{1\le k\le N\\ d\mid k}}\frac{c_k}{k}\Big)^2,\tag{11.6}$$

because $d\mid(j,k)$ exactly when $d\mid j$ and $d\mid k$. □

Corollary 11.3 (The first K factors)**.** *For every integer $K\ge2$,*

Reviewed by Claude (review 2)

$$\frac{\pi^2}{24}\Big(2-\frac{\pi^2}{6}\Big)K\le\operatorname{av}\Big[\Big(\sum_{k=1}^{K}h(x/k)\Big)^2\Big]\le\frac{\pi^2}{12}\Big(5K+(1+\log K)^2\Big).\tag{11.7}$$

Proof. For an integer $n\ge1$ let $\operatorname{har}(n)=\sum_{b=1}^{n}1/b$. For $1\le d\le K$ the integers $k\le K$ divisible by d are $k=db$ with $1\le b\le\lfloor K/d\rfloor$, so $\sum_{k\le K,\ d\mid k}1/k=\operatorname{har}(\lfloor K/d\rfloor)/d$. By (11.5) with $N=K$ and $c_k=1$,

$$\operatorname{av}\Big[\Big(\sum_{k=1}^{K}h(x/k)\Big)^2\Big]=\frac{\pi^2}{12}\sum_{d=1}^{K}\frac{J_2(d)}{d^2}\,\operatorname{har}(\lfloor K/d\rfloor)^2.\tag{11.8}$$

By (11.3), $0<J_2(d)/d^2\le1$. For the upper bound, $\operatorname{har}(n)\le1+\log n$ gives $\operatorname{har}(\lfloor K/d\rfloor)\le1+\log(K/d)$. The function $\tau\mapsto(1+\log(K/\tau))^2$ is decreasing on $[1,K]$, so

$$\sum_{d=1}^{K}\Big(1+\log\frac{K}{d}\Big)^2\le(1+\log K)^2+\int_1^K\Big(1+\log\frac{K}{\tau}\Big)^2d\tau\le(1+\log K)^2+K\int_0^1(1-\log u)^2\,du,\tag{11.9}$$

and $\int_0^1(1-\log u)^2\,du=1+2+2=5$. For the lower bound we keep only the terms with $K/2<d\le K$. For them $\lfloor K/d\rfloor=1$ and $\operatorname{har}(1)=1$. Since $\prod_b(1-\theta_b)\ge1-\sum_b\theta_b$ for numbers $\theta_b\in[0,1]$, (11.3) gives $J_2(d)/d^2\ge1-\sum_{n\ge2}1/n^2=2-\pi^2/6$. The number of integers d with $K/2<d\le K$ is $K-\lfloor K/2\rfloor\ge K/2$. This gives the lower bound in (11.7). □

12. The power weight on a half-line

In this section $\sigma>0$ is real, $j,k\ge1$ are integers, $g=(j,k)$, $j=gj'$ and $k=gk'$. We write $a=j\vee k$, $\omega=2\pi/(jk)$ and $w(x)=x^{-2\sigma-1}$. We study

$$I_{jk}(\sigma)=\int_{j\vee k}^{\infty}h(x/j)\,h(x/k)\,x^{-2\sigma-1}\,dx,\tag{12.1}$$

whose absolute convergence is part of Proposition 12.5.

Reviewed by Claude (review 2)

Definition 12.1 (Pair sums)**.** For $r \in \mathbb{Z}$ let

$$T_{jk}(r) = \sum_{\substack{m,n\geq 1 \\ mk-nj=r}} \frac{1}{2mn}, \tag{12.2}$$

a series of positive terms whose convergence is proved in Lemma 12.2. For integers $r \geq 1$ let

$$U_{jk}(r) = \sum_{\substack{m,n\geq 1 \\ mk+nj=r}} \frac{1}{2mn}, \qquad D_{jk}(r) = T_{jk}(r) + T_{jk}(-r) + U_{jk}(r). \tag{12.3}$$

The sum defining $U_{jk}(r)$ is finite, and it is 0 when it is empty.

Reviewed by Claude (review 2)

Lemma 12.2 (Closed form of the pair sums)**.** *Let $r \in \mathbb{Z}$. If $g \nmid r$, then $T_{jk}(r) = 0$. If $r = 0$, then*

$$T_{jk}(0) = \frac{\pi^2}{12} \cdot \frac{g^2}{jk}. \tag{12.4}$$

If $r = gr'$ with an integer $r' \neq 0$, the equation $mk' - nj' = r'$ has solutions in integers $m, n \geq 1$. Let (m_*, n_*) be the one with the least m. Then the series (12.2) *converges and*

$$T_{jk}(r) = \frac{1}{2r'}\Big(\psi\Big(\frac{m_*}{j'}\Big) - \psi\Big(\frac{n_*}{k'}\Big)\Big). \tag{12.5}$$

Proof. Since g divides $mk - nj$, the sum (12.2) is empty when $g \nmid r$. Let $r = gr'$. Then $mk - nj = r$ is the same as $mk' - nj' = r'$. Let $r' = 0$. Since $(j', k') = 1$, the equation $mk' = nj'$ with $m, n \geq 1$ forces $j' \mid m$, so the solutions are $m = j'l$, $n = k'l$ with $l \geq 1$, and $T_{jk}(0) = \sum_{l\geq 1} 1/(2j'k'l^2) = \pi^2/(12j'k')$, which is (12.4).

Let $r' \neq 0$. Since $(j', k') = 1$ there are integers m_0, n_0 with $m_0k' - n_0j' = r'$. If (m, n) is another integer solution, then $(m - m_0)k' = (n - n_0)j'$, so $j' \mid m - m_0$, and $(m, n) = (m_0 + j'l, n_0 + k'l)$ for an integer l. Conversely every such pair is a solution. Both coordinates increase strictly with l. Hence the solutions with $m, n \geq 1$ are exactly the pairs with $l \geq l_0$ for some integer l_0, that is, the pairs $(m_* + j'l, n_* + k'l)$ with $l \geq 0$, where (m_*, n_*) is the solution with $m, n \geq 1$ and m least.

Put $\alpha = m_*/j'$ and $\beta = n_*/k'$. Then $\alpha, \beta > 0$ and $\alpha - \beta = (m_*k' - n_*j')/(j'k') = r'/(j'k') \neq 0$. For $l \geq 0$,

$$\frac{1}{2(m_* + j'l)(n_* + k'l)} = \frac{1}{2j'k'(l+\alpha)(l+\beta)} = \frac{1}{2j'k'(\alpha-\beta)}\Big(\frac{1}{l+\beta} - \frac{1}{l+\alpha}\Big) = \frac{1}{2r'}\Big(\frac{1}{l+\beta} - \frac{1}{l+\alpha}\Big). \tag{12.6}$$

The left side is $O(l^{-2})$, so the series converges. Summing over $l \geq 0$ and using the formula for $\psi(\alpha) - \psi(\beta)$ stated after (2.3) gives (12.5). This holds for both signs of r'. Each term $1/(l+\beta) - 1/(l+\alpha)$ has the sign of $\alpha - \beta$, which is the sign of r'. Hence in (12.5) the difference $\psi(m_*/j') - \psi(n_*/k')$ and the denominator $2r'$ change sign together, and $T_{jk}(r) > 0$. □

Reviewed by Claude (review 2)

Lemma 12.3 (Fourier coefficients of a product of two factors)**.** *The function $F(x) = h(x/j)h(x/k)$ is even, has period jk, and is square integrable on $[0, jk]$. Its Fourier coefficients with respect to the period jk are*

$$\widehat{F}_{jk}(0) = T_{jk}(0), \qquad \widehat{F}_{jk}(r) = \widehat{F}_{jk}(-r) = \tfrac{1}{2}D_{jk}(r) \quad (r \geq 1). \tag{12.7}$$

Consequently $F(x) = T_{jk}(0) + \sum_{r\geq 1} D_{jk}(r)\cos(2\pi rx/(jk))$ in $L^2(0, jk)$, and

$$\sum_{r\geq 1} D_{jk}(r)^2 = 2\operatorname{av}\big[(F - T_{jk}(0))^2\big] < \infty. \tag{12.8}$$

Proof. Put $F_1(x) = h(x/j)$ and $F_2(x) = h(x/k)$. Both are even and have period jk. By the Cauchy–Schwarz inequality and Lemma 3.1 with $n = 4$, $\int_0^{jk} F^2 \le (\int_0^{jk} F_1^4)^{1/2}(\int_0^{jk} F_2^4)^{1/2} < \infty$.

By (3.1), $F_1(x) = \sum_{m\neq 0} \widehat{h}_1(m) e^{2\pi i (mk)x/(jk)}$ and $F_2(x) = \sum_{n\neq 0} \widehat{h}_1(n) e^{2\pi i (nj)x/(jk)}$ in $L^2(0, jk)$. Hence $\widehat{(F_1)}_{jk}(\mathfrak{a}) = \widehat{h}_1(m)$ if $\mathfrak{a} = mk$ with $m \neq 0$ and $\widehat{(F_1)}_{jk}(\mathfrak{a}) = 0$ otherwise, and $\widehat{(F_2)}_{jk}(\mathfrak{a}) = \widehat{h}_1(n)$ if $\mathfrak{a} = nj$ with $n \neq 0$ and $\widehat{(F_2)}_{jk}(\mathfrak{a}) = 0$ otherwise. Fix $\mathfrak{q} \in \mathbb{Z}$ and let $F_3(x) = F_2(x) e^{2\pi i \mathfrak{q} x/(jk)}$. Then $\widehat{(F_3)}_{jk}(\mathfrak{a}) = \widehat{(F_2)}_{jk}(\mathfrak{a} - \mathfrak{q})$, and $\overline{\widehat{(F_2)}_{jk}(\mathfrak{a} - \mathfrak{q})} = \widehat{(F_2)}_{jk}(\mathfrak{q} - \mathfrak{a})$ because F_2 is real. Since F_2 is real, $F(x) e^{-2\pi i \mathfrak{q} x/(jk)} = F_1(x)\overline{F_3(x)}$, and (2.2) with $\Phi_1 = F_1$, $\Phi_2 = F_3$ and $L = jk$ gives

$$\widehat{F}_{jk}(\mathfrak{q}) = \operatorname{av}[F_1 \overline{F_3}] = \sum_{\mathfrak{a}\in\mathbb{Z}} \widehat{(F_1)}_{jk}(\mathfrak{a})\, \widehat{(F_2)}_{jk}(\mathfrak{q} - \mathfrak{a}) = \sum_{\substack{m,n\neq 0 \\ mk+nj=\mathfrak{q}}} \frac{1}{4|m|\,|n|}, \tag{12.9}$$

with absolute convergence, where we used (3.1).

Let $\mathfrak{q} = r \ge 1$. The pairs with $m, n > 0$ contribute $\frac{1}{2}U_{jk}(r)$. The pairs with $m > 0 > n$ are $(m, -n_1)$ with $n_1 \ge 1$ and $mk - n_1 j = r$, and they contribute $\frac{1}{2}T_{jk}(r)$. The pairs with $m < 0 < n$ are $(-m_1, n)$ with $m_1 \ge 1$ and $m_1 k - nj = -r$, and they contribute $\frac{1}{2}T_{jk}(-r)$. There is no pair with $m, n < 0$. Hence $\widehat{F}_{jk}(r) = \frac{1}{2}D_{jk}(r)$. For $\mathfrak{q} = 0$ there is no pair with m, n of the same sign, and the two remaining classes contribute $\frac{1}{2}T_{jk}(0)$ each. Since F is real and even, $\widehat{F}_{jk}(-r) = \widehat{F}_{jk}(r)$. This proves (12.7). The cosine expansion follows, and (12.8) is (2.2) for $\Phi_1 = \Phi_2 = F - T_{jk}(0)$ with $L = jk$. □

Reviewed by Claude (review 2)

Lemma 12.4 (The kernel κ_σ)**.** *For real $y > 0$ put*

$$\kappa_\sigma(y) = 2\sigma\, y^{2\sigma} \int_y^\infty \frac{\cos v}{v^{2\sigma+1}}\, dv. \tag{12.10}$$

Then $|\kappa_\sigma(y)| \le \min(1, 4\sigma/y)$, and for all real $x > 0$ and $\lambda > 0$,

$$\int_x^\infty \cos(2\pi\lambda u)\, u^{-2\sigma-1}\, du = \frac{x^{-2\sigma}}{2\sigma}\, \kappa_\sigma(2\pi\lambda x). \tag{12.11}$$

Proof. Since $|\cos v| \le 1$ and $\int_y^\infty v^{-2\sigma-1} dv = y^{-2\sigma}/(2\sigma)$, we get $|\kappa_\sigma(y)| \le 1$. Integration by parts gives

$$\int_y^\infty \frac{\cos v}{v^{2\sigma+1}}\, dv = -\frac{\sin y}{y^{2\sigma+1}} + (2\sigma+1)\int_y^\infty \frac{\sin v}{v^{2\sigma+2}}\, dv, \tag{12.12}$$

whose modulus is at most $y^{-2\sigma-1} + y^{-2\sigma-1}$. Hence $|\kappa_\sigma(y)| \le 4\sigma/y$. The substitution $v = 2\pi\lambda u$ in the left side of (12.11) gives $(2\pi\lambda)^{2\sigma} \int_{2\pi\lambda x}^\infty \cos v\, v^{-2\sigma-1} dv$, which equals the right side by (12.10) with $y = 2\pi\lambda x$. □

Reviewed by Claude (review 2)

Proposition 12.5 (The gcd term and an explicit bound)**.** *The integral* (12.1) *converges absolutely, and*

$$\Big| I_{jk}(\sigma) - \frac{\pi^2}{12} \cdot \frac{g^2}{jk} \cdot \frac{(j \vee k)^{-2\sigma}}{2\sigma} \Big| \le \frac{\pi^2}{6}\, [j,k]\, (j \vee k)^{-2\sigma-1}. \tag{12.13}$$

Proof. Put $L = [j,k]$, $F(x) = h(x/j)h(x/k)$ and $F_0 = F - T_{jk}(0)$, where $T_{jk}(0)$ is the pair sum (12.2) with $r = 0$. By Theorem 11.1 and (12.4), F has period L and $\operatorname{av}[F_0] = 0$. By the Cauchy–Schwarz inequality and (11.1) with $j = k$, $\operatorname{av}[|F|] \le (\operatorname{av}[h(x/j)^2] \operatorname{av}[h(x/k)^2])^{1/2} = \pi^2/12$. Also $0 < T_{jk}(0) \le \pi^2/12$ because $g^2 \le jk$. Hence $\int_0^L |F_0| \le L\pi^2/6$.

The absolute convergence follows from $\int_a^\infty |F| x^{-2\sigma-1} dx \le \sum_{b\ge 0} (a + bL)^{-2\sigma-1} \int_0^L |F| < \infty$. For $x \ge a$ let $Y(x) = \int_a^x F_0$. The integral of F_0 over any interval of length L is 0, so $Y(x)$ equals the integral of F_0 over an interval of length less than L, and $|Y(x)| \le L\pi^2/6$. For $X > a$, integration

by parts gives

$$\int_a^X F_0(x)\,x^{-2\sigma-1}\,dx = Y(X)X^{-2\sigma-1} + (2\sigma+1)\int_a^X Y(x)\,x^{-2\sigma-2}\,dx. \tag{12.14}$$

Letting $X\to\infty$, the modulus of the limit is at most $(2\sigma+1)(L\pi^2/6)\int_a^\infty x^{-2\sigma-2}dx = (L\pi^2/6)a^{-2\sigma-1}$. Since $\int_a^\infty T_{jk}(0)x^{-2\sigma-1}dx = T_{jk}(0)a^{-2\sigma}/(2\sigma)$, this is (12.13). □

Reviewed by Claude (review 2)

Theorem 12.6 (Weighted Gram matrix)**.** *With the notation of* (12.1), (12.2), (12.3) *and* (12.10),

$$I_{jk}(\sigma) = \frac{(j\vee k)^{-2\sigma}}{2\sigma}\Big[\frac{\pi^2}{12}\cdot\frac{g^2}{jk} + \sum_{r\ge1} D_{jk}(r)\,\kappa_\sigma\Big(\frac{2\pi(j\vee k)\,r}{jk}\Big)\Big], \tag{12.15}$$

and the series converges absolutely.

Proof. Put $F(x) = h(x/j)h(x/k)$ and $F_0 = F - T_{jk}(0)$. By Lemma 12.4, $|\kappa_\sigma(\omega ar)| \le 4\sigma/(\omega ar)$, and by the Cauchy–Schwarz inequality and (12.8), $\sum_{r\ge1}|D_{jk}(r)|/r \le (\sum_r D_{jk}(r)^2)^{1/2}(\sum_r r^{-2})^{1/2} < \infty$. This proves the absolute convergence in (12.15).

Let $Y_0(x) = \int_0^x F_0$ for $x\in\mathbb{R}$. It is absolutely continuous, and it has period jk because $\mathrm{av}[F_0] = 0$. For an integer $\mathfrak{q}\neq0$, integration by parts over one period gives $\widehat{(Y_0)}_{jk}(\mathfrak{q}) = \widehat{(F_0)}_{jk}(\mathfrak{q})/(i\omega\mathfrak{q})$, which by (12.7) equals $D_{jk}(|\mathfrak{q}|)/(2i\omega\mathfrak{q})$. Define

$$Y_1(x) = \sum_{r\ge1}\frac{D_{jk}(r)}{\omega r}\sin(\omega rx). \tag{12.16}$$

The series converges absolutely and uniformly because $\sum_r |D_{jk}(r)|/r < \infty$, so Y_1 is continuous with period jk, and its Fourier coefficient at $\mathfrak{q}\neq0$ is $D_{jk}(|\mathfrak{q}|)/(2i\omega\mathfrak{q})$ as well. Thus $Y_0 - Y_1$ is a continuous function with period jk whose Fourier coefficients at all $\mathfrak{q}\neq0$ vanish. By the remark after (2.2) it is constant, and the constant is $Y_0(0) - Y_1(0) = 0$. Hence $Y_0 = Y_1$.

Since Y_0 is bounded and $w(X)\to0$, integration by parts gives

$$\int_a^\infty F_0(x)\,w(x)\,dx = -Y_0(a)\,w(a) - \int_a^\infty Y_0(x)\,w'(x)\,dx. \tag{12.17}$$

Insert $Y_0 = Y_1$ and (12.16). The series converges uniformly and w' is integrable on $[a,\infty)$, so the right side of (12.17) is the sum over $r\ge1$ of

$$\frac{D_{jk}(r)}{\omega r}\Big(-\sin(\omega ra)\,w(a) - \int_a^\infty \sin(\omega rx)\,w'(x)\,dx\Big) = D_{jk}(r)\int_a^\infty \cos(\omega rx)\,w(x)\,dx, \tag{12.18}$$

where the equality is one more integration by parts. By (12.11) with $x=a$ and $\lambda = r/(jk)$, the right side of (12.18) equals $D_{jk}(r)\,a^{-2\sigma}\kappa_\sigma(\omega ar)/(2\sigma)$. Finally $I_{jk}(\sigma) = T_{jk}(0)\int_a^\infty w + \int_a^\infty F_0 w$, with $\int_a^\infty w = a^{-2\sigma}/(2\sigma)$, and (12.4) gives (12.15). □

By Lemma 12.2, $D_{jk}(r) = 0$ unless $g \mid r$, because g also divides $mk+nj$. The first term in the bracket of (12.15) is the entry of the gcd matrix of Theorem 11.1. It comes from the pairs of harmonics with $mk = nj$. The series collects the pairs with $mk\neq nj$, each weighted by κ_σ.

13. Numerical comparison

Table 3 compares the two sides of (12.15) in four cases. It is a numerical check and is not used in any proof. The left side $I_{jk}(\sigma)$ was computed by numerical integration of (12.1) over 150 periods of length $[j,k]$, with the remaining tail replaced by its gcd term. By (12.13), applied on the tail, the error of this replacement is at most $\frac{\pi^2}{6}[j,k]X^{-2\sigma-1}$ with $X = j\vee k + 150\,[j,k]$. In the four cases of the table this bound equals $4.1\cdot10^{-7}$, $1.5\cdot10^{-7}$, $2.5\cdot10^{-5}$ and $1.4\cdot10^{-4}$, rounded up. The right side was computed with the terms $r\le20000$ of the series, using (12.5) for $T_{jk}(\pm r)$. The error of the quadrature and the tail $r>20000$ of the series are not bounded here. The last column is the gcd

term $\frac{\pi^2}{12}\frac{g^2}{jk}\frac{(j\vee k)^{-2\sigma}}{2\sigma}$ alone. The two computed values differ by at most $1.3\cdot 10^{-8}$ in the four cases. For $(j,k)=(2,3)$ and $(j,k)=(4,6)$ the weighted entry is negative.

TABLE 3. The two sides of (12.15), and the gcd term alone.

j	k	σ	integral (12.1)	series (12.15)	gcd term
2	3	0.75	−0.00447151	−0.00447151	0.01758709
4	6	0.75	−0.00158092	−0.00158092	0.00621798
3	3	0.5	0.29787858	0.29787859	0.27415568
2	5	0.3	0.02940918	0.02940917	0.05218975

14. FIGURES

All the figures were computed from the definitions of this note. The function G was evaluated through (5.2) and (5.1), with a quadrature rule for the integral over x (the midpoint rule for Figure 2, a Gauss–Legendre rule with 240 nodes for the surfaces) and the Euler–Maclaurin formula for the tail of the sum over n. Figure 1 shows h on one period and the kernel $\mathcal{K}$ of (4.1). Figure 2 shows G on the real axis and illustrates Theorem 6.1.

For $0<\operatorname{Re}s<2$ let

$$A(s)=\frac{\log(2\pi)}{s}+\frac{1}{s^2}-\int_0^1 \log\frac{\sin(\pi u)}{\pi u}\,u^{-s-1}\,du. \tag{14.1}$$

The integral converges absolutely because $\log(\sin(\pi u)/(\pi u))=O(u^2)$ as $u\to 0$ and the function has a logarithmic singularity at $u=1$. Since $\int_1^\infty \log(2\pi u)u^{-s-1}du=\log(2\pi)/s+1/s^2$ for $\operatorname{Re}s>0$, and $h(u)-\log(2\pi u)=\log(\sin(\pi u)/(\pi u))$ for $0<u<1$, the definitions (1.2) and (14.1) give, for $0<\operatorname{Re}s<2$,

$$G(s)-A(s)=\int_0^\infty \big(h(u)-\log(2\pi u)\big)\,u^{-s-1}\,du. \tag{14.2}$$

Thus $G-A$ is the Mellin transform on the whole half-line of h minus its logarithmic part.

Reviewed by Claude (review 1)

Proposition 14.1 (Continuation and real zeros of A)**.** *Put $\Sigma_{2k}=\sum_{m\ge1}m^{-2k}$ for the integers $k\ge1$. The function A of* (14.1) *extends to a meromorphic function on $\mathbb{C}$, given by*

$$A(s)=\frac{\log(2\pi)}{s}+\frac{1}{s^2}+\sum_{k\ge1}\frac{\Sigma_{2k}}{k\,(2k-s)}. \tag{14.3}$$

Its poles are a double pole at 0 and a simple pole at each positive even integer. For $\operatorname{Re}s<0$,

$$A(s)=-\int_0^1 h(u)\,u^{-s-1}\,du=\frac{(s+1)(s+2)}{4\pi^2}\,\widetilde{H}(s),\qquad \widetilde{H}(s)=\int_0^1\mathcal{K}(u)\,u^{-s-3}\,du. \tag{14.4}$$

For real s we have $A(s)>0$ for $s<-2$, $A(-2)=0$, $A(s)<0$ for $-2<s<-1$, $A(-1)=0$, and $A(s)>0$ for $-1<s<0$ and for $0<s<2$. The zeros at -2 and -1 are simple. For every integer $l\ge1$ the function A has at least one zero in the interval $(2l,2l+2)$.

Proof. Let $0<u<1$. The classical product of Euler, $\sin(\pi u)=\pi u\prod_{m\ge1}(1-u^2/m^2)$, and the series $\log(1-\lambda)=-\sum_{k\ge1}\lambda^k/k$ for $0\le\lambda<1$ give

$$-\log\frac{\sin(\pi u)}{\pi u}=\sum_{m\ge1}\sum_{k\ge1}\frac{u^{2k}}{k\,m^{2k}}=\sum_{k\ge1}\frac{\Sigma_{2k}}{k}\,u^{2k}, \tag{14.5}$$

where the terms are nonnegative, so that the order of summation is free. Let $\sigma = \operatorname{Re} s < 2$. We have $\int_0^1 u^{2k-\sigma-1}du = 1/(2k-\sigma)$ for all $k \geq 1$, and $\sum_{k\geq 1} \Sigma_{2k}/(k(2k-\sigma)) < \infty$ because $\Sigma_{2k} \leq \Sigma_2$. Hence (14.5) can be integrated termwise against u^{-s-1}, and

$$-\int_0^1 \log \frac{\sin(\pi u)}{\pi u}\, u^{-s-1}\, du = \sum_{k\geq 1} \frac{\Sigma_{2k}}{k\,(2k-s)} \qquad (\operatorname{Re} s < 2). \tag{14.6}$$

With (14.1) this is (14.3) for $0 < \operatorname{Re} s < 2$. Let $\theta > 0$. For $|s| \leq \theta$ and $k > \theta$ we have $|2k - s| \geq 2k - \theta \geq k$, so the term of index k in (14.3) has modulus at most Σ_2/k^2. Hence the series, without its finitely many terms with $k \leq \theta$, converges uniformly on $|s| \leq \theta$. The right side of (14.3) is therefore meromorphic on $\mathbb{C}$, with the principal part $\log(2\pi)/s + 1/s^2$ at 0 and a simple pole of residue $-\Sigma_{2l}/l$ at $2l$ for every integer $l \geq 1$. It is the extension of A.

Let $\operatorname{Re} s < 0$. Then $\int_0^1 u^{-s-1}du = -1/s$ and $\int_0^1 (\log u)u^{-s-1}du = -1/s^2$, so $\int_0^1 \log(2\pi u)u^{-s-1}du = -\log(2\pi)/s - 1/s^2$. Since $h(u) = \log(2\pi u) + \log(\sin(\pi u)/(\pi u))$ for $0 < u < 1$, (14.3) and (14.6) give the first equality in (14.4). The integral converges absolutely by the bound $|h(u)| \leq \log 2 + |\log(4u)|$ for $0 < u \leq \frac{1}{2}$ from the proof of Lemma 3.1 and by $h(1-u) = h(u)$.

Let $0 < \tau < \frac{1}{4}$. By (4.3), $h = -\mathcal{K}''/(4\pi^2)$ almost everywhere, with $\mathcal{K}'$ absolutely continuous. Two integrations by parts, in which $\mathcal{K}'(1) = \mathcal{K}(1) = 0$ by (4.2), give

$$\int_\tau^1 h(u)\, u^{-s-1}\, du = \frac{\mathcal{K}'(\tau)\,\tau^{-s-1}}{4\pi^2} + \frac{(s+1)\,\mathcal{K}(\tau)\,\tau^{-s-2}}{4\pi^2} - \frac{(s+1)(s+2)}{4\pi^2}\int_\tau^1 \mathcal{K}(u)\, u^{-s-3}\, du. \tag{14.7}$$

By (4.3) and the bound for $|h|$ just quoted, $|\mathcal{K}'(\tau)| \leq 4\pi^2 \int_0^\tau |h| \leq 4\pi^2\tau(\log 2 + |\log(4\tau)| + 1)$. By Lemma 4.2, $0 \leq \mathcal{K}(\tau) \leq 2\pi^2\tau^2(\log\frac{1}{\tau} + \frac{3}{2}) + \pi^4\tau^4/9$. Since $\sigma < 0$, the two boundary terms in (14.7) have modulus at most a constant times $\tau^{-\sigma}(1 + \log\frac{1}{\tau})$ and tend to 0 as $\tau \to 0$, and $\mathcal{K}(u)u^{-\sigma-3}$ is integrable on $(0,1)$. Letting $\tau \to 0$ in (14.7) gives the second equality in (14.4), and $\widetilde{H}$ is holomorphic in $\operatorname{Re} s < 0$.

For real $s < 0$ we have $\widetilde{H}(s) > 0$, because $\mathcal{K} > 0$ on $(0,1)$ by (4.2). By (14.4) the sign of $A(s)$ is the sign of $(s+1)(s+2)$, and the zeros at -2 and -1 are simple. For $0 < s < 2$ every term of (14.3) is positive. Let $l \geq 1$ be an integer. On the interval $(2l, 2l+2)$ the function A is real and continuous. As s decreases to $2l$, the term of index l in (14.3) tends to $-\infty$ and the rest of the right side of (14.3) stays bounded. As s increases to $2l+2$, the term of index $l+1$ tends to $+\infty$ and the rest of the right side stays bounded. By the intermediate value theorem A vanishes in $(2l, 2l+2)$. □

Formula (14.4) is the counterpart of (5.2) for the part of $\mathcal{K}$ on $(0,1)$.

Figures 3, 4 and 6 show the moduli of G, A and $G - A$ on the rectangle $0.05 \leq \operatorname{Re} s \leq 1.95$, $1 \leq \operatorname{Im} s \leq 40$, on a linear scale. The vertical axis is inverted, with the value 0 at the top, so that a zero is a peak whose tip is at 0, where the red points are. The grid has 253×889 points. Its steps are 0.01 in $\operatorname{Re} s$ and 0.05 in $\operatorname{Im} s$, and it is refined near the six zeros of G of Table 2 and near the six points $\frac{1}{2} + it$ of Figure 8, which are grid points. In Figure 6 the values above 0.6 are clipped. At the six zeros of G the computed values of $|G|$ are between $4.1 \cdot 10^{-12}$ and $2.8 \cdot 10^{-11}$. On this rectangle the surface of $|A|$ does not reach 0. Figure 5 shows $|A(s)|$ in the same way for $-3.5 \leq \operatorname{Re} s \leq 9.5$ and $|\operatorname{Im} s| \leq 2$, computed from (14.3) and clipped above 1.5. Its peaks are the real zeros of Proposition 14.1 at -2 and -1 and the computed zeros 3.0428, 5.0981, 7.1478 and 9.1892, and its pits are the poles at $0, 2, 4, 6, 8$. Figure 7 shows in the same way, on the grid of Figure 3, the modulus of the Riemann zeta function $\zeta(s)$, clipped above 1.5, which is used elsewhere only in Sections 17 and 18. On that grid the computed values satisfy $|G(s) - A(s) - \frac{\pi}{s}\cot(\frac{\pi s}{2})\zeta(s)| \leq 1.2 \cdot 10^{-11}$. This relation is proved in Proposition 17.1.

Figure 8 shows, in the same rectangle, the six zeros of G of Table 2, the six points $\frac{1}{2} + it$ where t is the imaginary part of one of the first six nontrivial zeros of ζ, and the arcs of the curve $\operatorname{Im}(G/A) = 0$ on which $0 < \operatorname{Re}(G/A) < 1$. The quotient G/A is formed only at points where $A(s) \neq 0$. At the

253×889 grid points the computed values satisfy $|A(s)| \geq 0.072$, and the arcs are the level lines drawn from the grid values. That A has no zero for $0.05 \leq \operatorname{Re} s \leq 1$ and $9 \leq \operatorname{Im} s \leq 40$ is Theorem 15.4. On the rest of the rectangle it is not proved in this note. At those six points the computed value of $|G - A|$ is at most $1.1 \cdot 10^{-11}$. In the computed figure each arc joins a zero of G, where $G/A = 0$, to one of those points, where $G/A = 1$. This pairing is a numerical observation for these six zeros. It is not proved, and no correspondence between the zeros of G and the points with $G = A$ is claimed. Figure 12 shows the Gram matrices of Theorems 11.1 and 12.6 for $1 \leq j, k \leq 10$. In the second panel the entries are $2\sigma(j \vee k)^{2\sigma} I_{jk}(\sigma)$ with $\sigma = \frac{3}{4}$, that is, the bracket of (12.15), computed with the terms $r \leq 1500$. Of its 100 entries, 52 are negative.

The last three figures concern exponential sums over computed zeros. In the rectangle $-1.5 \leq \operatorname{Re} s \leq 3$, $0.2 \leq \operatorname{Im} s \leq 200$ a numerical search finds $n_G = 79$ zeros $z_1^G, \dots, z_{n_G}^G$ of G, and the computed variation of the argument of G along the boundary of this rectangle corresponds to 79 zeros. Their real parts lie between 0.1036 and 1.1274, and the six with the smallest imaginary parts are the zeros of Table 2. For A the same search finds no zero with $-60 \leq \operatorname{Re} s \leq 201$ and $0.2 \leq \operatorname{Im} s \leq 200$, and exactly one zero in each interval $(2l, 2l+2)$ with $1 \leq l \leq 99$. With -2 and -1 this gives $n_A = 101$ real zeros $z_1^A, \dots, z_{n_A}^A$ of A in $[-60, 200]$. These counts are computed and not proved. For real x put

$$\mathcal{E}_G(x) = \frac{1}{n_G} \sum_{b=1}^{n_G} e^{ix \operatorname{Im} z_b^G}, \qquad \mathcal{E}_A(x) = \frac{1}{n_A} \sum_{b=1}^{n_A} e^{ix z_b^A}. \tag{14.8}$$

For G the frequencies are the imaginary parts of the zeros. For A the zeros found are real, so their imaginary parts vanish, and the frequencies are the zeros themselves. Figure 9 shows the two curves $x \mapsto \mathcal{E}_G(x)$ and $x \mapsto \mathcal{E}_A(x)$ for $0 \leq x \leq 2\pi$, with x on the vertical axis. Figure 10 shows the same curves in the complex plane, and the two sets of zeros with their centroids, which are $\frac{1}{n_G} \sum_b z_b^G = 0.45923 + 119.21486\, i$ and $\frac{1}{n_A} \sum_b z_b^A = 99.44250$. The means $\frac{1}{2\pi} \int_0^{2\pi} \mathcal{E}_G(x)\, dx = -0.00029 + 0.00188\, i$ and $\frac{1}{2\pi} \int_0^{2\pi} \mathcal{E}_A(x)\, dx = 0.00151 + 0.00423\, i$ are the dashed lines of Figure 9.

Figure 11 shows the moduli. The mean of $|\mathcal{E}_G(x)|$ over $0 \leq x \leq 2\pi$ is 0.0847. For the integers $2 \leq n \leq 31$, the value $|\mathcal{E}_G(\log n)|$ lies between 0.152 and 0.254 for the eleven primes, between 0.090 and 0.189 for the prime powers $4, 8, 9, 16, 25, 27$, and between 0.020 and 0.124 for the other thirteen integers. At the eleven primes $p \leq 31$ the real part of $\mathcal{E}_G(\log p)$ is negative, between -0.161 and -0.030. The modulus $|\mathcal{E}_A(x)|$ equals 0.926 at $x = \pi$ and 0.784 at $x = 2\pi$, and it is below 0.074 for $0.3 \leq x \leq 2\pi - 0.2$ with $|x - \pi| \geq 0.2$. The zeros z_b^A other than -2 and -1 lie in consecutive intervals of length 2. All the statements of the last two paragraphs are numerical observations for the listed zeros.

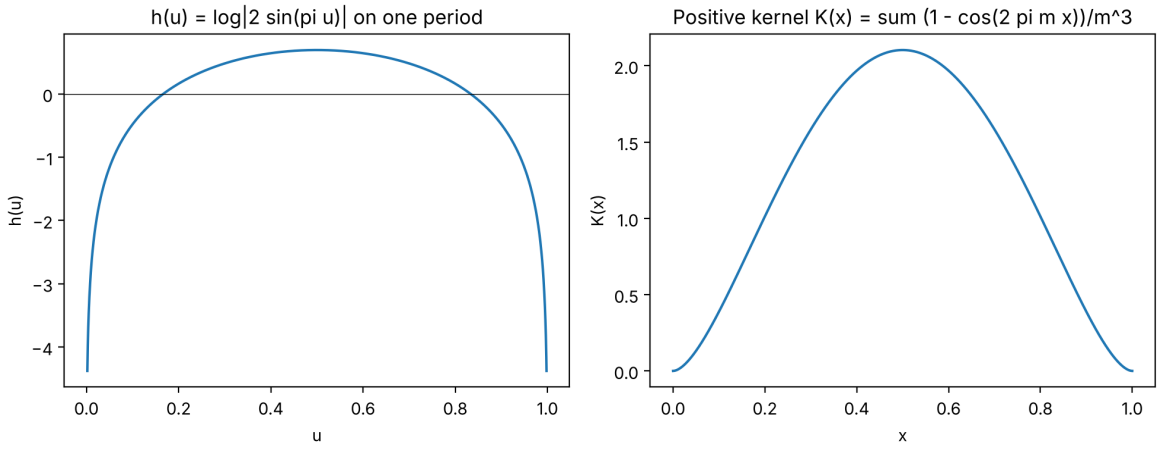


FIGURE 1. The function h on one period and the positive kernel $\mathcal{K}$.

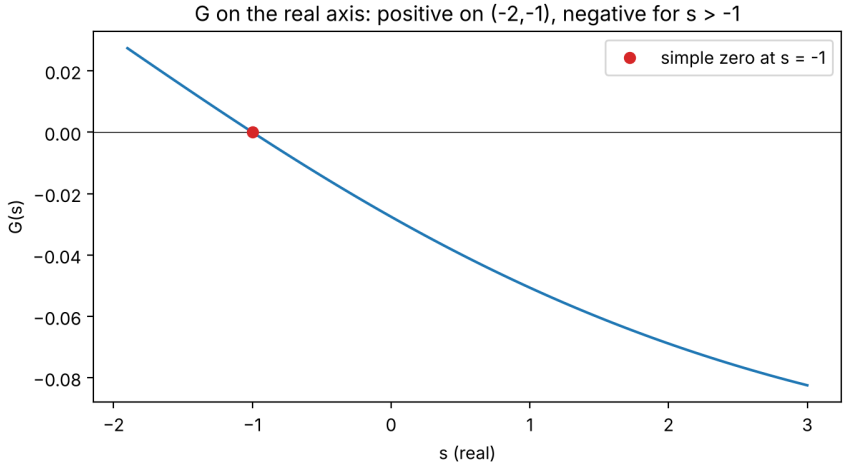


FIGURE 2. G on the real axis.

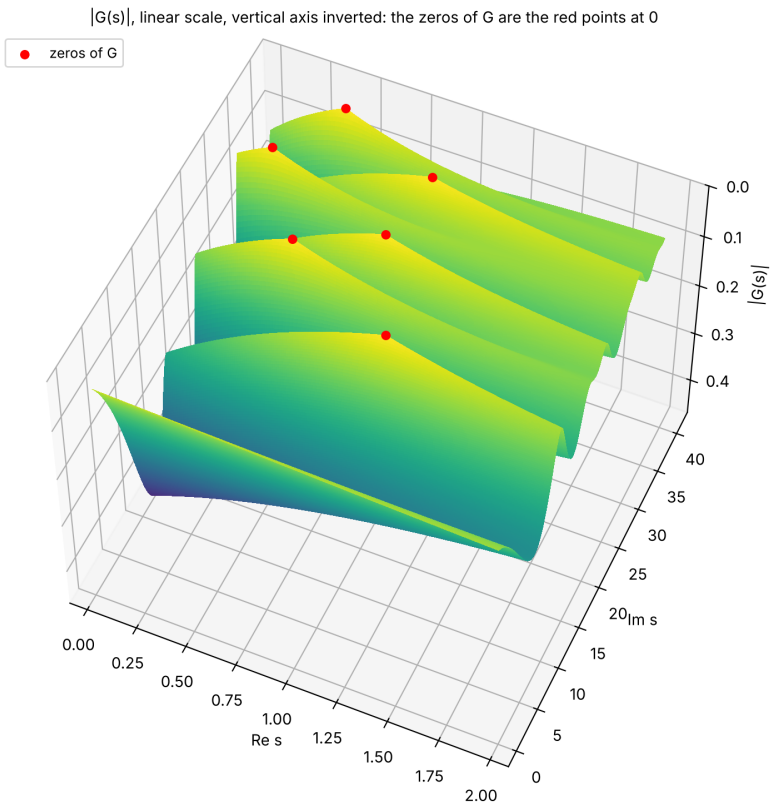


FIGURE 3. $|G(s)|$ on a linear scale, with the vertical axis inverted. The red points, at the value 0, are the six zeros of G of Table 2.

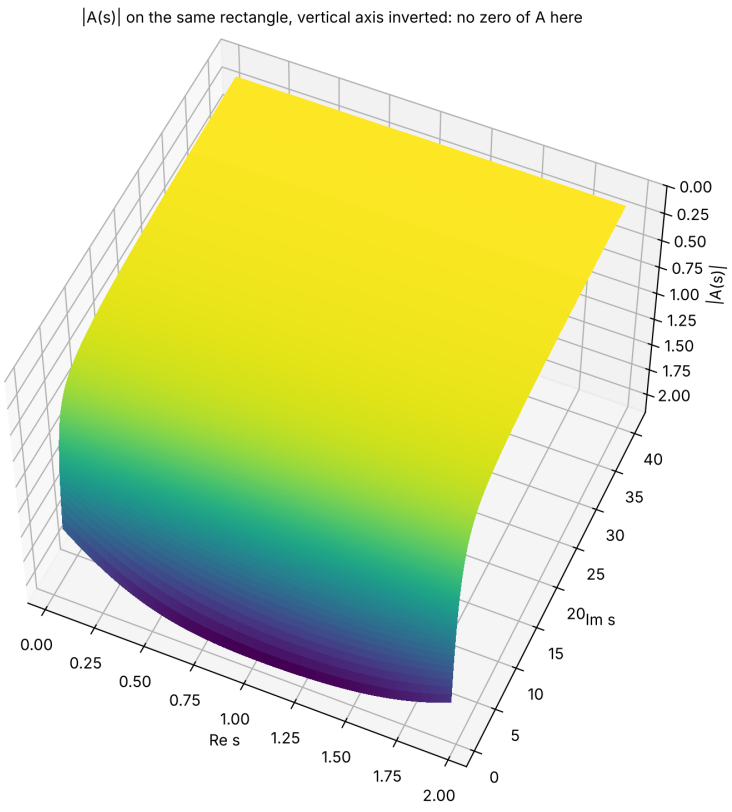


FIGURE 4. $|A(s)|$ on the rectangle of Figure 3, with the vertical axis inverted.

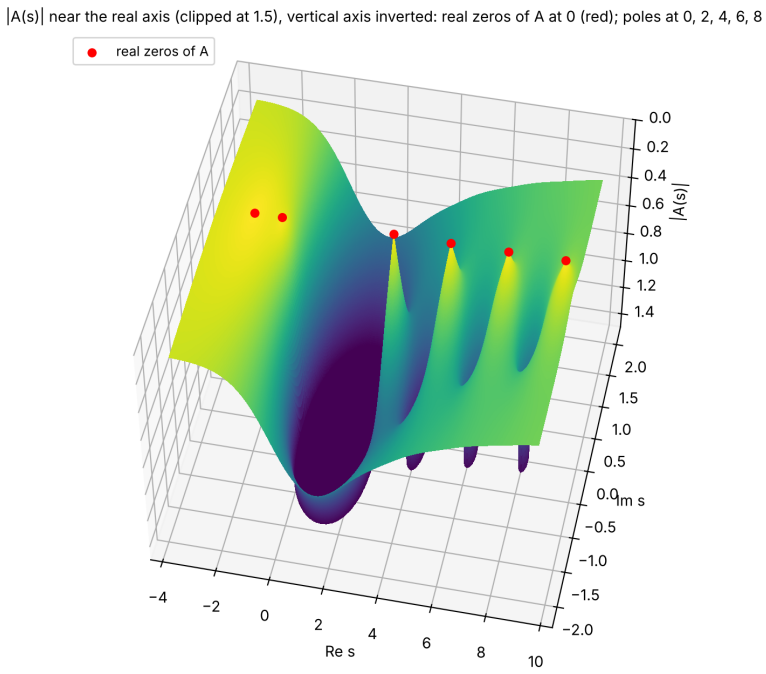


FIGURE 5. $|A(s)|$ near the real axis, clipped above 1.5, with the vertical axis inverted. The red points, at the value 0, are real zeros of A. The pits are its poles at $0, 2, 4, 6, 8$.

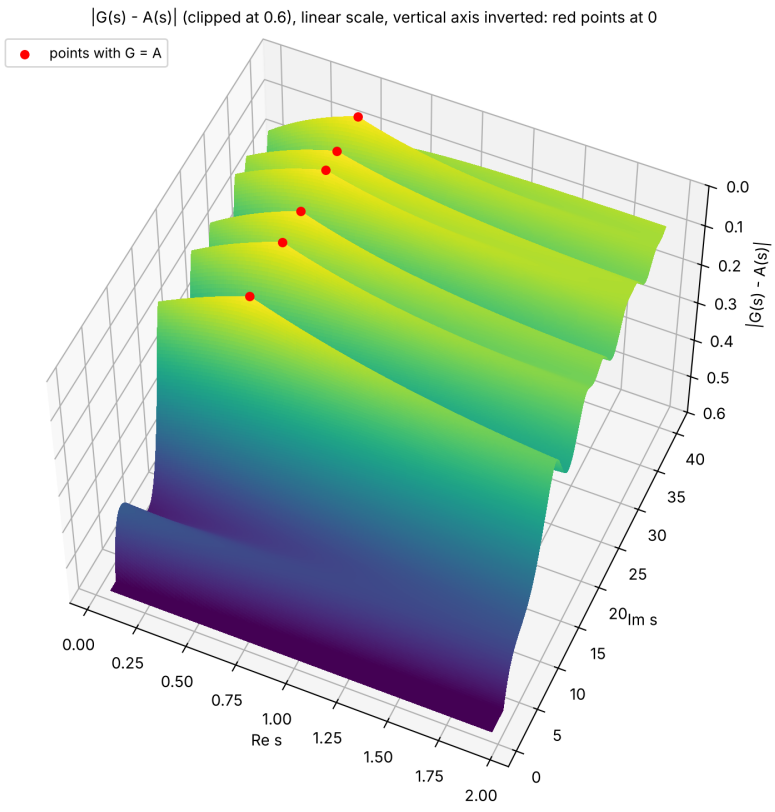


FIGURE 6. $|G(s) - A(s)|$, clipped above 0.6, with the vertical axis inverted. The red points, at the value 0, are the six points $\frac{1}{2} + it$ of Figure 8.

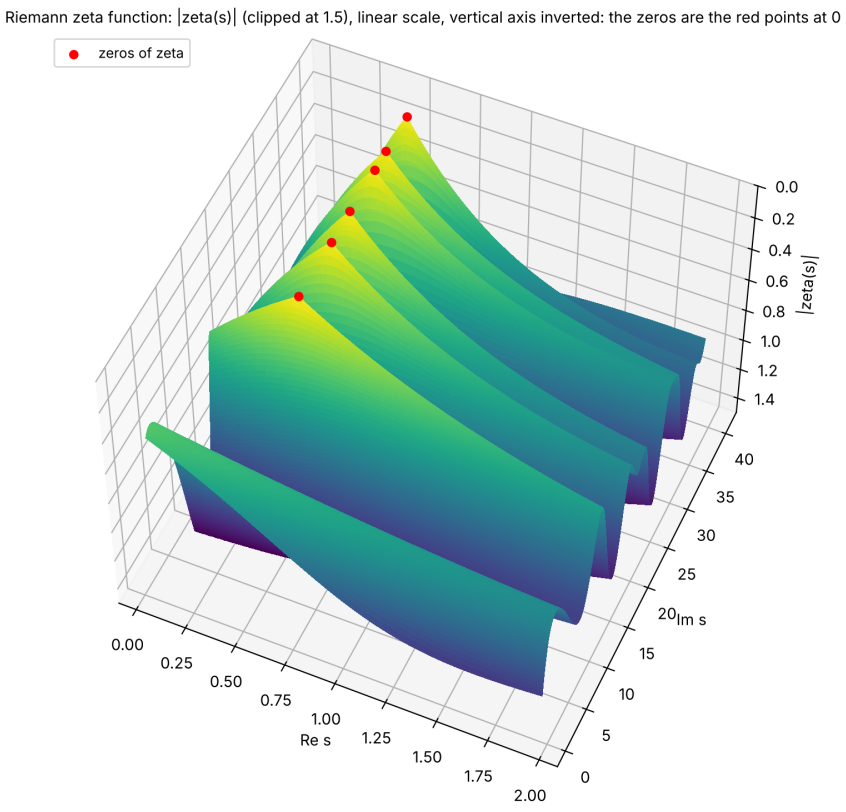


FIGURE 7. $|\zeta(s)|$, clipped above 1.5, with the vertical axis inverted. The red points, at the value 0, are the first six nontrivial zeros of ζ.

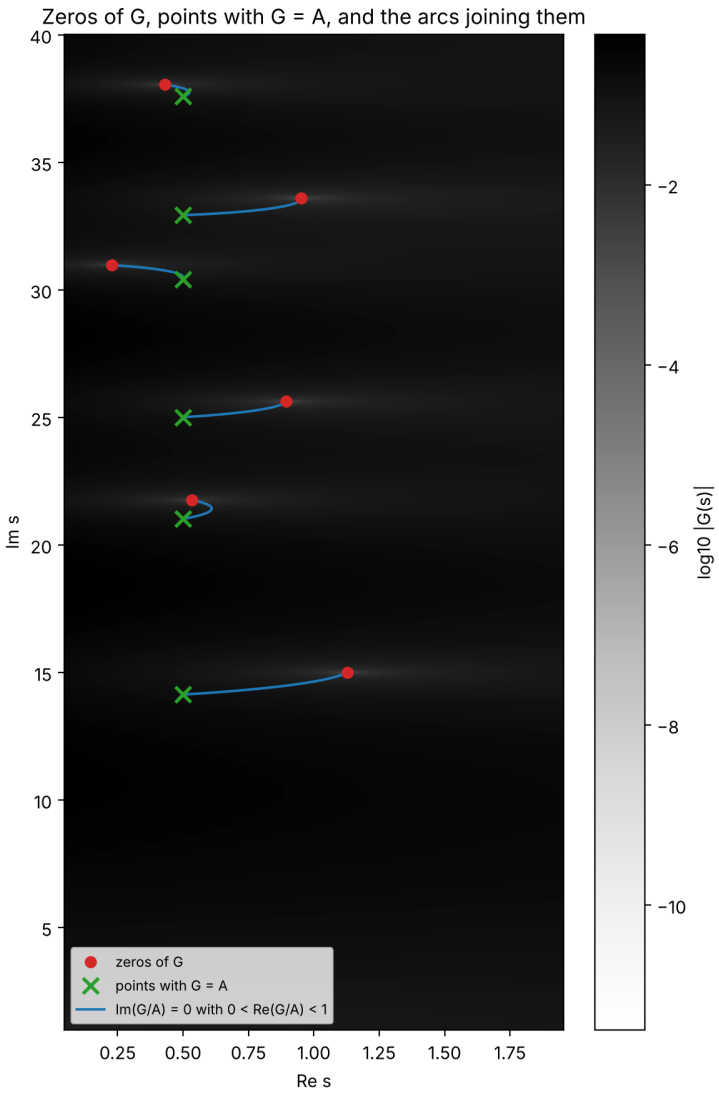


FIGURE 8. Computed zeros of G, computed points with $G = A$, and the computed arcs of $\mathrm{Im}(G/A) = 0$ with $0 < \mathrm{Re}(G/A) < 1$, at points where $A \neq 0$. The pairing shown by the arcs is a numerical observation.

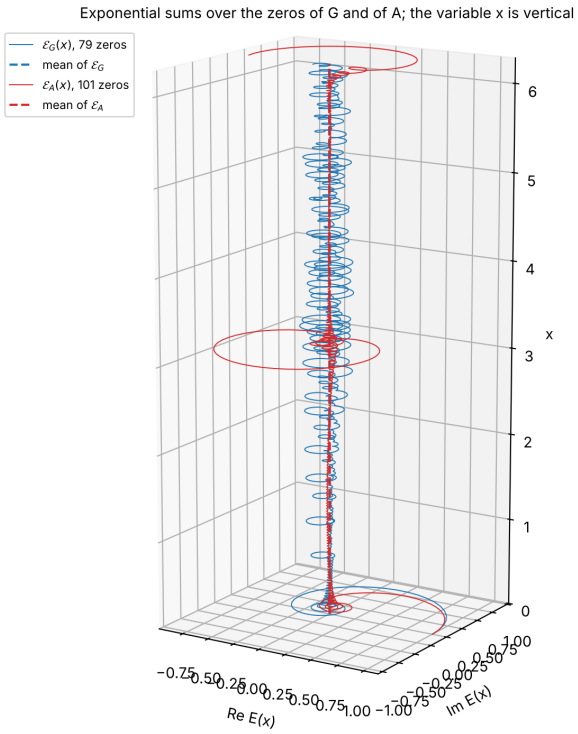


FIGURE 9. The exponential sums $\mathcal{E}_G(x)$ and $\mathcal{E}_A(x)$ of (14.8) for $0 \le x \le 2\pi$. The variable x is the vertical axis. The dashed lines are the means of the two curves.

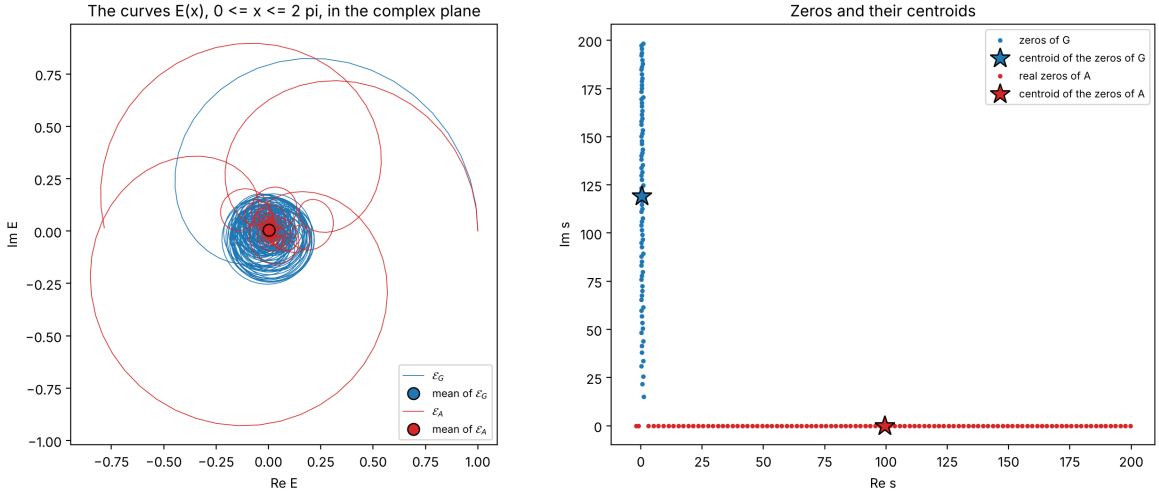


FIGURE 10. Left: the curves $\mathcal{E}_G$ and $\mathcal{E}_A$ in the complex plane. Right: the 79 computed zeros of G and the 101 computed real zeros of A, with their centroids.

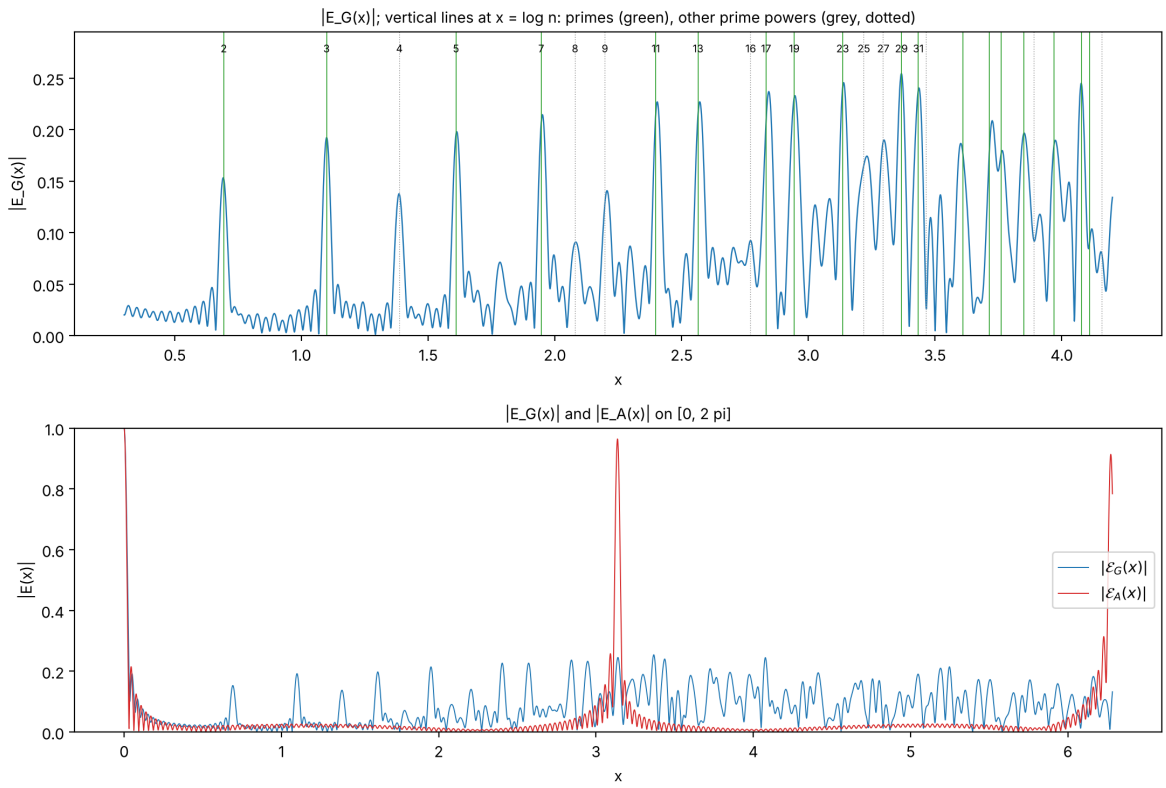


FIGURE 11. Top: $|\mathcal{E}_G(x)|$ for $0.3 \le x \le 4.2$, with vertical lines at $x = \log n$ for the prime powers $n \le 66$. Bottom: $|\mathcal{E}_G(x)|$ and $|\mathcal{E}_A(x)|$ for $0 \le x \le 2\pi$.

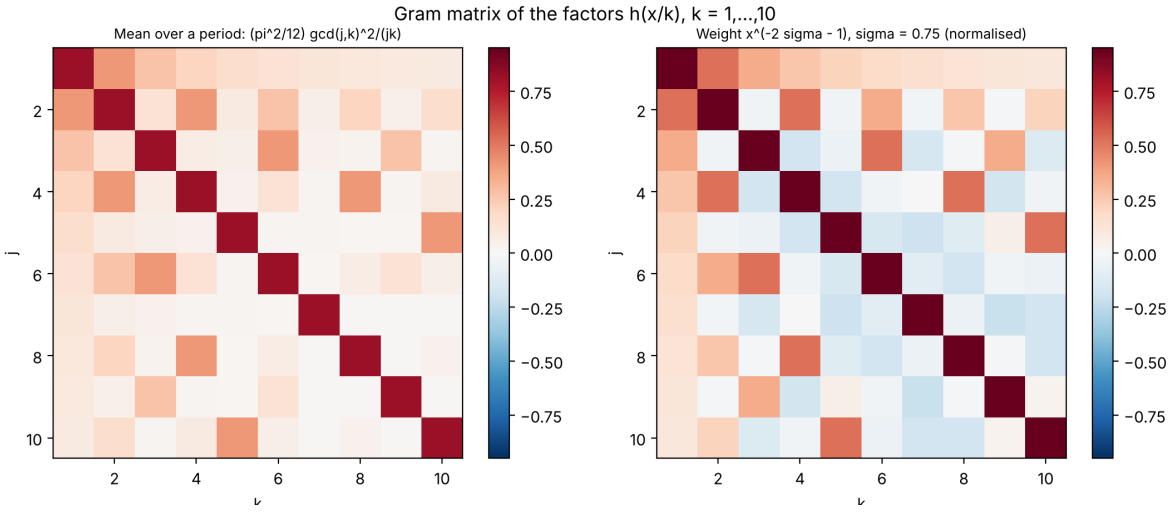


FIGURE 12. Gram matrices of the factors $h(x/k)$, $1 \le k \le 10$: mean over a period, and weight $x^{-2\sigma-1}$ with $\sigma = \frac{3}{4}$.

15. The difference $G - A$

This section and the next one use only the function h and the functions defined from it in the previous sections. The Riemann zeta function is not used. For real $u > 0$ with $u \notin \mathbb{Z}$ put

$$\ell(u) = \log \frac{\pi u}{|\sin(\pi u)|}. \tag{15.1}$$

Since $|\sin(\pi u)| < \pi u$ for $u > 0$, we have $\ell(u) > 0$. By (1.1), $h(u) - \log(2\pi u) = -\ell(u)$, so (14.2) reads

$$G(s) - A(s) = -\int_0^\infty \ell(u)\, u^{-s-1}\, du \qquad (0 < \operatorname{Re} s < 2). \tag{15.2}$$

Proposition 15.1 (Sign and size of $G - A$)**.** *Let $0 < \sigma < 2$. Then $G(\sigma) - A(\sigma) < 0$, and for every real t* Reviewed by Claude (review 1)

$$|G(\sigma + it) - A(\sigma + it)| \le A(\sigma) - G(\sigma), \tag{15.3}$$

with equality only for $t = 0$. In particular $G(s) \neq A(s)$ for real s with $0 < s < 2$.

Proof. The integral in (15.2) converges absolutely, as stated after (14.1). The substitution $u = e^v$ gives $G(\sigma + it) - A(\sigma + it) = -\int_{\mathbb{R}} f(v) e^{-itv}\, dv$ with $f(v) = \ell(e^v) e^{-\sigma v}$, and $f(v) > 0$ for every v with $e^v \notin \mathbb{Z}$. Hence $G(\sigma) - A(\sigma) = -\int_{\mathbb{R}} f < 0$ and $|G(\sigma + it) - A(\sigma + it)| \le \int_{\mathbb{R}} f = A(\sigma) - G(\sigma)$. Suppose that equality holds, and choose a real θ with $\int_{\mathbb{R}} f(v) e^{-itv} dv = e^{i\theta} \int_{\mathbb{R}} f$. Then $\int_{\mathbb{R}} f(v)(1 - \cos(tv + \theta)) dv = 0$. The integrand is nonnegative and $f > 0$ almost everywhere, so $\cos(tv + \theta) = 1$ for almost every real v, which forces $t = 0$. □

For $0 < u < 1$ put

$$q_0(u) = \log \frac{\pi u\,(1 - u^2)}{\sin(\pi u)} = \sum_{k \ge 1} \frac{\Sigma_{2k} - 1}{k}\, u^{2k}, \qquad Q_0(s) = \int_0^1 q_0(u)\, u^{-s-1}\, du \quad (\operatorname{Re} s < 2). \tag{15.4}$$

The series follows from (14.5) and $\log(1 - u^2) = -\sum_{k \ge 1} u^{2k}/k$. Its terms are nonnegative, and $\Sigma_{2k} - 1 = \sum_{m \ge 2} m^{-2k} \le 4^{-k} + \int_2^\infty \lambda^{-2k} d\lambda = 4^{-k}(1 + \frac{2}{2k-1}) \le 3 \cdot 4^{-k}$. Hence $0 \le q_0(u) \le 3 \sum_{k \ge 1} 4^{-k}/k = 3 \log \frac{4}{3}$ on $(0, 1)$, and termwise integration gives

$$Q_0(s) = \sum_{k \ge 1} \frac{\Sigma_{2k} - 1}{k\,(2k - s)} \qquad (\operatorname{Re} s < 2). \tag{15.5}$$

The right side of (15.5) is meromorphic on $\mathbb{C}$, with simple poles at the even integers $2k \ge 2$, and we use it as the definition of Q_0 on $\mathbb{C}$.

Proposition 15.2 (The difference $G - A$ through the digamma function)**.** *For every complex $s \neq 0$ that is not an even integer,* Reviewed by Claude (review 1)

$$sA(s) = \log(2\pi) + \frac{1}{s} - \psi\Big(1 - \frac{s}{2}\Big) - \gamma + s\, Q_0(s) \qquad (\operatorname{Re} s < 2), \tag{15.6}$$

and for $0 < \operatorname{Re} s < 2$, with Q as in (7.1)*,*

$$G(s) - A(s) = \frac{\psi(1 - \frac{s}{2}) - \psi(1 + s)}{s} - Q(s) - Q_0(s), \tag{15.7}$$

where

$$\frac{\psi(1 - \frac{s}{2}) - \psi(1 + s)}{s} = -\sum_{n \ge 1} \frac{3}{(n + s)(2n - s)}. \tag{15.8}$$

For $0 < \operatorname{Re} s < 2$ the real part of the series in (15.8) *is positive, so the real part of the first term of* (15.7) *is negative.*

Proof. Let $\operatorname{Re} s < 2$, $s \neq 0$ and $s \notin 2\mathbb{Z}$. For $k \geq 1$ we have $1/(k(2k-s)) = \frac{1}{s}(1/(k-\frac{s}{2}) - 1/k)$. By (2.3) with argument $1-\frac{s}{2}$, whose real part is positive,

$$\sum_{k\geq 1} \frac{1}{k\,(2k-s)} = -\frac{1}{s}\sum_{l\geq 0}\Big(\frac{1}{l+1} - \frac{1}{l+1-\frac{s}{2}}\Big) = -\frac{\psi(1-\frac{s}{2})+\gamma}{s}. \tag{15.9}$$

Writing $\Sigma_{2k} = 1 + (\Sigma_{2k}-1)$ in (14.3) and using (15.9) and (15.5) gives (15.6). For $0 < \operatorname{Re} s < 2$, (7.2) gives $G(s) = (\log(2\pi) - \gamma - \psi(s))/s - Q(s)$. Subtracting (15.6) divided by s, and using $\psi(1+s) = \psi(s) + 1/s$, which follows from (2.3), gives (15.7). By (2.3) again,

$$\psi\Big(1-\frac{s}{2}\Big) - \psi(1+s) = \sum_{n\geq 1}\Big(\frac{1}{n+s} - \frac{1}{n-\frac{s}{2}}\Big) = -\sum_{n\geq 1}\frac{3s}{(n+s)(2n-s)}, \tag{15.10}$$

which is (15.8). Write $s = \sigma + it$. Then $\operatorname{Re}((n+s)(2n-s)) = 2n^2 + n\sigma - \sigma^2 + t^2 \geq 2 + \sigma - \sigma^2 + t^2 = (2-\sigma)(1+\sigma) + t^2 > 0$ for $n \geq 1$ and $0 < \sigma < 2$. Since $\operatorname{Re}(1/w) = \operatorname{Re} w/|w|^2$ for $w \neq 0$, every term of the series has positive real part. □

The next lemma and theorem give an analytic proof that A does not vanish in the closed strip $0 \leq \operatorname{Re} s \leq 1$ above height 9.

Reviewed by Claude (review 1)

Lemma 15.3 (A lower bound for the real part of the digamma function)**.** *For real x with $\frac{1}{2} \leq x \leq 1$ and real y,*

$$\operatorname{Re}\psi(x+iy) \geq \frac{1}{2}\log(x^2+y^2) - \gamma - \log 2 + \frac{y^2}{2x\,(x^2+y^2)}. \tag{15.11}$$

Proof. By (2.3), $\operatorname{Re}\psi(x+iy) - \psi(x) = \sum_{l\geq 0} f(l)$ with $f(\lambda) = y^2/((\lambda+x)((\lambda+x)^2+y^2))$ for $\lambda \geq 0$. With $w = \lambda + x > 0$ one finds

$$\frac{d^2}{dw^2}\,\frac{1}{w\,(w^2+y^2)} = \frac{12w^4 + 6w^2y^2 + 2y^4}{w^3(w^2+y^2)^3} > 0, \tag{15.12}$$

so f is convex. For a convex function the chord lies above the graph, so $\int_l^{l+1} f \leq \frac{1}{2}(f(l) + f(l+1))$ for every integer $l \geq 0$. Summing over l, and using $f(\lambda) \to 0$ as $\lambda \to \infty$, gives $\int_0^\infty f \leq \sum_{l\geq 0} f(l) - \frac{1}{2}f(0)$. Since $\int_0^\infty f = \int_x^\infty y^2\,dw/(w(w^2+y^2)) = \frac{1}{2}\log(1+y^2/x^2)$, we obtain $\operatorname{Re}\psi(x+iy) \geq \psi(x) + \frac{1}{2}\log(1+y^2/x^2) + y^2/(2x(x^2+y^2))$. Termwise differentiation of (2.3) for real $x > 0$ gives $\frac{d}{dx}(\psi(x) - \log x) = \sum_{l\geq 0}(l+x)^{-2} - \frac{1}{x} > 0$, because $(l+x)^{-2} > \int_l^{l+1}(\lambda+x)^{-2}d\lambda$ and the integrals add up to $\frac{1}{x}$. By (2.3), $\psi(\frac{1}{2}) = -\gamma - 2\sum_{l\geq 0}(\frac{1}{2l+1} - \frac{1}{2l+2}) = -\gamma - 2\log 2$. Hence for $x \geq \frac{1}{2}$ we have $\psi(x) \geq \log x + \psi(\frac{1}{2}) + \log 2 = \log x - \gamma - \log 2$, and $\log x + \frac{1}{2}\log(1+y^2/x^2) = \frac{1}{2}\log(x^2+y^2)$ gives (15.11). □

Reviewed by Claude (review 1)

Theorem 15.4 (A does not vanish above height 9)**.** *Let $0 \leq \sigma \leq 1$ and $|t| \geq 9$, and put*

$$\mathcal{Q}(t) = \log(8\pi) - \frac{1}{2}\log(1+t^2) - \frac{t^2}{2(t^2+4)} + \frac{1}{t^2} - 0.64\,\frac{t^2-1}{t^2+4} + 10^{-27}. \tag{15.13}$$

Then $\operatorname{Re}(sA(s)) \leq \mathcal{Q}(t) < 0$ for $s = \sigma + it$. In particular $A(s) \neq 0$.

Proof. Since $A(\bar{s}) = \overline{A(s)}$ by (14.3), we may assume $t \geq 9$. We bound the real parts of the terms of (15.6). First, $\operatorname{Re}(1/s) = \sigma/(\sigma^2+t^2) \leq 1/t^2$. Second, apply Lemma 15.3 with $x = 1 - \frac{\sigma}{2} \in [\frac{1}{2}, 1]$ and $y = -\frac{t}{2}$. Then $x^2+y^2 \geq \frac{1}{4}(1+t^2)$, and $x \leq 1$, $x^2+y^2 \leq \frac{1}{4}(4+t^2)$ give $y^2/(2x(x^2+y^2)) \geq t^2/(2(t^2+4))$. Hence

$$-\operatorname{Re}\psi\Big(1-\frac{s}{2}\Big) - \gamma \leq 2\log 2 - \frac{1}{2}\log(1+t^2) - \frac{t^2}{2(t^2+4)}. \tag{15.14}$$

Third, by (15.5), $\mathrm{Re}(sQ_0(s)) = \sum_{k\ge1} \frac{\Sigma_{2k}-1}{k}\,\mathfrak{r}_k$ with

$$\mathfrak{r}_k = \mathrm{Re}\,\frac{s}{2k-s} = \frac{2k\sigma-\sigma^2-t^2}{(2k-\sigma)^2+t^2}. \tag{15.15}$$

For $0\le\sigma\le1$ we have $2k\sigma-\sigma^2\le 2k-1$, because the left side increases with σ on $[0,1]$. For $k=1$ the numerator is at most $1-t^2<0$ and the denominator is at most $4+t^2$, so $\rho_1\le -(t^2-1)/(t^2+4)$. If $2\le k$ and $2k-1<t^2$, then $\mathfrak{r}_k<0$. If $2k-1\ge t^2$, then $k\ge41$ because $t^2\ge81$, and $\mathfrak{r}_k\le(2k-1)/(2k-1)^2=1/(2k-1)$, since the denominator is at least $(2k-1)^2$. With $\Sigma_{2k}-1\le 3\cdot4^{-k}$ these terms add up to at most $\frac{3}{41\cdot81}\sum_{k\ge41}4^{-k}=4^{-40}/3321<10^{-27}$. Finally $\Sigma_2-1>\sum_{m=2}^{10}m^{-2}+\int_{11}^{\infty}\lambda^{-2}d\lambda>0.5497+0.0909>0.64$. Adding up, $\mathrm{Re}(sA(s))\le\mathcal{Q}(t)$.

Each of the terms $-\frac12\log(1+t^2)$, $-\frac{t^2}{2(t^2+4)}=-\frac12+\frac{2}{t^2+4}$, $\frac{1}{t^2}$ and $-0.64\frac{t^2-1}{t^2+4}=-0.64+\frac{3.2}{t^2+4}$ decreases for $t>0$, so $\mathcal{Q}(t)\le\mathcal{Q}(9)$ for $t\ge9$. Since $e^{3.2242}>25.1328>8\pi$ and $e^{4.4066}<82$, we have $\log(8\pi)<3.2242$ and $\frac12\log82>2.2033$. With $\frac{81}{170}>0.4764$, $\frac{1}{81}<0.0124$ and $0.64\cdot\frac{80}{85}>0.6023$ this gives $\mathcal{Q}(9)<3.2242-2.2033-0.4764+0.0124-0.6023+10^{-27}<-0.045$. □

Remark 15.5 (A computer-assisted complement). Theorem 15.4 leaves the rectangle $0\le\mathrm{Re}\,s\le1$, $|\mathrm{Im}\,s|<9$, and its proof gives nothing there. On the real segment $0<s<2$ we have $A(s)>0$ by Proposition 14.1. The program `A_nonvanishing_check.py` of the files that accompany this note encloses $s^2A(s)$, written through (15.6), in ball arithmetic [1] on 188892 boxes that cover $0\le\mathrm{Re}\,s\le1$, $0\le\mathrm{Im}\,s\le102$, and none of the enclosures contains 0. This is a computer-assisted verification of $A(s)\ne0$ on that rectangle. No other statement of this note depends on it.

16. A positive convex kernel for $G-A$

Put $\mathfrak{b}=\log(2\pi)-\frac32$. Since $2\pi>e^{3/2}$, we have $\mathfrak{b}>0$. For real $u\ge0$ put

$$\mathcal{B}(u)=4\pi^2\int_0^u(u-\lambda)\,\ell(\lambda)\,d\lambda,\qquad \Omega(s)=\int_0^\infty\mathcal{B}(u)\,u^{-s-3}\,du\quad(0<\mathrm{Re}\,s<2). \tag{16.1}$$

The integral defining $\mathcal{B}$ converges because ℓ is nonnegative and locally integrable on $[0,\infty)$: it has only logarithmic singularities at the positive integers, and $\ell(\lambda)\le\pi^2\lambda^2/3$ for $0<\lambda\le\frac12$ by the bound $|\log(\sin(\pi\tau)/(\pi\tau))|\le\pi^2\tau^2/3$ from the proof of Lemma 4.2.

Reviewed by Claude (review 1)

Proposition 16.1 (The kernel $\mathcal{B}$)**.** *The function $\mathcal{B}$ has the following properties.* (i) *$\mathcal{B}$ is continuously differentiable on $[0,\infty)$, $\mathcal{B}'(u)=4\pi^2\int_0^u\ell$ is absolutely continuous and nondecreasing, $\mathcal{B}''=4\pi^2\ell$ almost everywhere, $\mathcal{B}$ is convex, and $\mathcal{B}(u)>0$ for $u>0$.* (ii) *For every $u>0$,*

$$\mathcal{B}(u)=\mathcal{K}(u)+2\pi^2u^2(\log u+\mathfrak{b}). \tag{16.2}$$

In particular $\mathcal{B}(1)=2\pi^2\mathfrak{b}$ and $\mathcal{B}'(1)=2\pi^2(1+2\mathfrak{b})$. (iii) *For $0\le u\le1$,*

$$\mathcal{B}(u)=4\pi^2\sum_{k\ge1}\frac{\Sigma_{2k}}{k\,(2k+1)(2k+2)}\,u^{2k+2}, \tag{16.3}$$

and $0<\mathcal{B}(u)<2\pi^2\mathfrak{b}\,u^4$ for $0<u<1$. (iv) *The integral $\Omega(s)$ converges absolutely for $0<\mathrm{Re}\,s<2$, and there*

$$G(s)-A(s)=-\frac{(s+1)(s+2)}{4\pi^2}\,\Omega(s). \tag{16.4}$$

Proof. (i) By Fubini's theorem $\int_0^u(u-\lambda)\ell(\lambda)d\lambda=\int_0^u(\int_0^\mu\ell)d\mu$. The inner integral is a continuous nondecreasing function of μ, absolutely continuous with derivative ℓ almost everywhere. This gives the statements on $\mathcal{B}'$ and $\mathcal{B}''$. A function with nondecreasing derivative is convex, and $\mathcal{B}(u)>0$ for $u>0$ because $\ell>0$.

(ii) By (4.3), $\mathcal{K}(0)=0$ and Fubini's theorem, $\mathcal{K}(u)=\int_0^u\mathcal{K}'(\mu)\,d\mu=-4\pi^2\int_0^u\int_0^\mu h(\lambda)\,d\lambda\,d\mu=-4\pi^2\int_0^u(u-\lambda)h(\lambda)\,d\lambda$. Since $\ell(\lambda)+h(\lambda)=\log(2\pi\lambda)$ and $\int_0^u(u-\lambda)\log(2\pi\lambda)\,d\lambda=\frac{u^2}{2}\log(2\pi u)-\frac{3u^2}{4}$,

we get $\mathcal{B}(u)-\mathcal{K}(u)=2\pi^2u^2(\log(2\pi u)-\frac{3}{2})$, which is (16.2). At $u=1$, (4.2) gives $\mathcal{K}(1)=\mathcal{K}'(1)=0$, hence $\mathcal{B}(1)=2\pi^2\mathfrak{b}$ and $\mathcal{B}'(1)=2\pi^2(2\cdot1\cdot\log1+1+2\mathfrak{b})=2\pi^2(1+2\mathfrak{b})$.

(iii) For $0\le\lambda<1$, (14.5) gives $\ell(\lambda)=\sum_{k\ge1}\Sigma_{2k}\lambda^{2k}/k$ with nonnegative terms. Since $\int_0^u(u-\lambda)\lambda^{2k}d\lambda=u^{2k+2}/((2k+1)(2k+2))$, monotone convergence gives (16.3) for $0\le u\le1$. All the coefficients are positive and all the exponents are at least 4. Hence for $0<u<1$ every term with $k\ge2$ is strictly smaller than its value at $u=1$ times u^4, the term with $k=1$ equals its value at $u=1$ times u^4, and $0<\mathcal{B}(u)<u^4\mathcal{B}(1)=2\pi^2\mathfrak{b}\,u^4$.

(iv) Let $\sigma=\operatorname{Re}s$. Near 0 we have $\mathcal{B}(u)u^{-\sigma-3}<2\pi^2\mathfrak{b}\,u^{1-\sigma}$, which is integrable for $\sigma<2$. For $u\ge1$, (4.2) and (16.2) give $\mathcal{B}(u)\le\frac{5}{2}+2\pi^2u^2(\log u+\mathfrak{b})$, and $\mathcal{B}(u)u^{-\sigma-3}$ is integrable on $[1,\infty)$ for $\sigma>0$. Let $0<\tau<\frac12<X$. Since $\mathcal{B}'$ is absolutely continuous with $\mathcal{B}''=4\pi^2\ell$, two integrations by parts give

$$4\pi^2\int_\tau^X\ell(u)\,u^{-s-1}\,du=\Big[\mathcal{B}'(u)u^{-s-1}+(s+1)\mathcal{B}(u)u^{-s-2}\Big]_\tau^X+(s+1)(s+2)\int_\tau^X\mathcal{B}(u)\,u^{-s-3}\,du. \tag{16.5}$$

For $u\le\frac12$ the bound $\ell(\lambda)\le\pi^2\lambda^2/3$ gives $\mathcal{B}'(u)\le4\pi^4u^3/9$ and $\mathcal{B}(u)\le\pi^4u^4/9$, so the terms at τ are $O(\tau^{2-\sigma})$ and tend to 0. The series of derivatives in the proof of Proposition 4.1 gives $|\mathcal{K}'|\le2\pi\Sigma_2$, so by (16.2) $\mathcal{B}'(X)=O(X\log X)$ and $\mathcal{B}(X)=O(X^2\log X)$, and the terms at X are $O(X^{-\sigma}\log X)$ and tend to 0. Letting $\tau\to0$ and $X\to\infty$ and using (15.2) gives (16.4). □

By (16.4), and since $(s+1)(s+2)\ne0$ for $0<\operatorname{Re}s<2$, the solutions of $G(s)=A(s)$ in the strip $0<\operatorname{Re}s<2$ are exactly the zeros of Ω there.

Reviewed by Claude (review 1)

Proposition 16.2 (An inversion inequality and two positive weights)**.** *For $u>1$ and $0<\sigma\le1$,*

$$\mathcal{B}(u)>u^{2\sigma+4}\,\mathcal{B}(1/u). \tag{16.6}$$

For real σ with $0<\sigma<2$ and real v put

$$\upsilon_\sigma(v)=e^{-(\sigma+2)v}\mathcal{B}(e^v),\qquad \upsilon_\sigma^\pm(v)=\upsilon_\sigma(v)\pm\upsilon_\sigma(-v)\quad(v>0). \tag{16.7}$$

Then $\Omega(\sigma+it)=\int_{\mathbb{R}}\upsilon_\sigma(v)e^{-itv}\,dv$ for real t, and

$$\operatorname{Re}\Omega(\sigma+it)=\int_0^\infty\upsilon_\sigma^+(v)\cos(tv)\,dv,\qquad \operatorname{Im}\Omega(\sigma+it)=-\int_0^\infty\upsilon_\sigma^-(v)\sin(tv)\,dv. \tag{16.8}$$

For $0<\sigma\le1$ and $v>0$ we have $\upsilon_\sigma^+(v)>\upsilon_\sigma^-(v)>0$. The function Ω is positive on the real interval $(0,2)$ and strictly decreasing on $(0,1]$.

Proof. Let $u>1$ and $0<\sigma\le1$. By Proposition 16.1(iii) with $1/u$ in place of u, $u^{2\sigma+4}\mathcal{B}(1/u)<2\pi^2\mathfrak{b}\,u^{2\sigma}\le2\pi^2\mathfrak{b}\,u^2$. By (16.2) and $\mathcal{K}\ge0$, $\mathcal{B}(u)\ge2\pi^2u^2(\log u+\mathfrak{b})>2\pi^2\mathfrak{b}\,u^2$. This proves (16.6). The substitution $u=e^v$ in (16.1) gives the integral over $\mathbb{R}$. Splitting it at $v=0$ and replacing v by $-v$ on $(-\infty,0)$ gives (16.8). For $v>0$, (16.6) with $u=e^v$ is $\upsilon_\sigma(v)>\upsilon_\sigma(-v)$, and $\upsilon_\sigma(-v)>0$ by Proposition 16.1(i). For real $\sigma\in(0,2)$, $\Omega(\sigma)=\int_{\mathbb{R}}\upsilon_\sigma>0$. On a compact interval $[\sigma_1,\sigma_2]\subset(0,2)$ we have $|v|\upsilon_\sigma(v)\le|v|(\upsilon_{\sigma_1}(v)+\upsilon_{\sigma_2}(v))$, which is integrable, so differentiation under the integral sign is allowed and $\Omega'(\sigma)=-\int_{\mathbb{R}}v\,\upsilon_\sigma(v)\,dv=-\int_0^\infty v\,\upsilon_\sigma^-(v)\,dv$. For $0<\sigma\le1$ this is negative. □

Reviewed by Claude (review 1)

Theorem 16.3 (A region without solutions of $G=A$ near the real axis)**.** *For $0<\sigma<2$ and $j=0,1,2$ put $\mathcal{M}_j(\sigma)=\int_0^\infty(\log u)^j\,\mathcal{B}(u)\,u^{-\sigma-3}\,du$. If t is real and*

$$t^2(\mathcal{M}_0(\sigma)\mathcal{M}_2(\sigma)-\mathcal{M}_1(\sigma)^2)<2\,\mathcal{M}_0(\sigma)^2, \tag{16.9}$$

then $G(\sigma+it)\ne A(\sigma+it)$. In particular $G(\sigma+it)\ne A(\sigma+it)$ whenever $|t|<\sqrt{2\mathcal{M}_0(\sigma)/\mathcal{M}_2(\sigma)}$.

Proof. By the substitution $u=e^v$, $\mathcal{M}_j(\sigma)=\int_{\mathbb{R}}v^j\upsilon_\sigma(v)\,dv$, and these integrals converge by Proposition 16.1(iii) and (iv). Put $\mu=\mathcal{M}_1/\mathcal{M}_0$. By Proposition 16.2, $e^{it\mu}\Omega(\sigma+it)=\int_{\mathbb{R}}\upsilon_\sigma(v)e^{-it(v-\mu)}dv$.

Its real part is $\int_{\mathbb{R}} \upsilon_\sigma(v)\cos(t(v-\mu))\,dv$, and $\cos y \ge 1 - \frac{y^2}{2}$ for real y gives

$$|\Omega(\sigma+it)| \ge \mathcal{M}_0 - \frac{t^2}{2}\int_{\mathbb{R}} (v-\mu)^2 \upsilon_\sigma(v)\,dv = \mathcal{M}_0 - \frac{t^2}{2}\Big(\mathcal{M}_2 - \frac{\mathcal{M}_1^2}{\mathcal{M}_0}\Big). \tag{16.10}$$

Under (16.9) the right side is positive, and (16.4) gives $G \ne A$. The last statement follows because $\mathcal{M}_0\mathcal{M}_2 - \mathcal{M}_1^2 \le \mathcal{M}_0\mathcal{M}_2$. □

Reviewed by Claude (review 1)

Proposition 16.4 (Decay of Ω)**.** *Let $0 < \sigma < 2$. The function υ_σ is continuously differentiable on $\mathbb{R}$, υ_σ' is absolutely continuous, υ_σ'' is integrable on $\mathbb{R}$, and for real $t \ne 0$*

$$|\Omega(\sigma+it)| \le \frac{1}{t^2}\int_{\mathbb{R}} |\upsilon_\sigma''(v)|\,dv. \tag{16.11}$$

Proof. Put $u = e^v$. By Proposition 16.1(i), $\upsilon_\sigma'(v) = u^{-\sigma-2}\big(u\mathcal{B}'(u) - (\sigma+2)\mathcal{B}(u)\big)$ is continuous and absolutely continuous, and almost everywhere $\upsilon_\sigma''(v) = u^{-\sigma-2}(4\pi^2u^2\ell(u) - (2\sigma+3)u\mathcal{B}'(u) + (\sigma+2)^2\mathcal{B}(u))$. With $dv = du/u$, the three terms contribute to $\int_{\mathbb{R}}|\upsilon_\sigma''|$ at most constant multiples of $\int_0^\infty \ell(u)u^{-\sigma-1}du$, $\int_0^\infty \mathcal{B}'(u)u^{-\sigma-2}du$ and $\int_0^\infty \mathcal{B}(u)u^{-\sigma-3}du$. The first is finite by (15.2), and the other two by the bounds for $\mathcal{B}$ and $\mathcal{B}'$ near 0 and near ∞ in the proof of Proposition 16.1(iv). The same bounds show that $\upsilon_\sigma(v)$ and $\upsilon_\sigma'(v)$ tend to 0 as $v \to \pm\infty$. Two integrations by parts in $\Omega(\sigma+it) = \int_{\mathbb{R}} \upsilon_\sigma(v)e^{-itv}dv$ give $\Omega(\sigma+it) = -t^{-2}\int_{\mathbb{R}} \upsilon_\sigma''(v)e^{-itv}dv$, and (16.11) follows. □

The inequality (16.6) holds for every $\sigma \in (0,1]$, so Proposition 16.2 does not distinguish the value $\sigma = \frac12$. The next proposition shows that positivity and periodicity of the kernel alone cannot place the solutions of $G = A$ on $\operatorname{Re} s = \frac12$. For real x put

$$\mathcal{K}^\sharp(x) = 1 - \cos(2\pi x) + 2^{-11/4}(1 - \cos(4\pi x)). \tag{16.12}$$

Like $\mathcal{K}$, the function $\mathcal{K}^\sharp$ is even, has period 1, satisfies $\mathcal{K}^\sharp > 0$ off the integers and $\mathcal{K}^\sharp(n) = (\mathcal{K}^\sharp)'(n) = 0$ at the integers n. Put $H^\sharp(s) = \int_1^\infty \mathcal{K}^\sharp(u)u^{-s-3}du$ for $\operatorname{Re} s > -2$, $\widetilde{H}^\sharp(s) = \int_0^1 \mathcal{K}^\sharp(u)u^{-s-3}du$ for $\operatorname{Re} s < 0$, and

$$\mathcal{C}(s) = \int_0^\infty (1-\cos y)\, y^{-s-3}\,dy \qquad (-2 < \operatorname{Re} s < 0). \tag{16.13}$$

In view of (5.2) and (14.4), the counterparts of G and A for $\mathcal{K}^\sharp$ are $G^\sharp = -\frac{(s+1)(s+2)}{4\pi^2}H^\sharp$ and $A^\sharp = \frac{(s+1)(s+2)}{4\pi^2}\widetilde{H}^\sharp$.

Reviewed by Claude (review 1)

Proposition 16.5 (A positive kernel with solutions off the line)**.** *The functions $\widetilde{H}^\sharp$ and $\mathcal{C}$ extend to meromorphic functions on $\operatorname{Re} s > -2$ whose poles are real, and on $\operatorname{Re} s > -2$*

$$G^\sharp(s) - A^\sharp(s) = -\frac{(s+1)(s+2)}{4\pi^2}\,(2\pi)^{s+2}\big(1 + 2^{s-3/4}\big)\,\mathcal{C}(s). \tag{16.14}$$

Consequently $G^\sharp(s) = A^\sharp(s)$ at the points $s = \frac34 + i(2k+1)\pi/\log 2$, $k \in \mathbb{Z}$, which have real part $\frac34$ and nonzero imaginary part.

Proof. The Taylor series $\mathcal{K}^\sharp(u) = \sum_{n\ge1} \mathfrak{c}_n u^{2n}$ at 0 converges for all u, with $|\mathfrak{c}_n| \le (4\pi)^{2n}/(2n)!$. Termwise integration gives $\widetilde{H}^\sharp(s) = \sum_{n\ge1} \mathfrak{c}_n/(2n-2-s)$ for $\operatorname{Re} s < 0$, and the right side is meromorphic on $\mathbb{C}$ with poles at the even integers $2n-2 \ge 0$. In the same way $\int_0^1 (1-\cos y)y^{-s-3}dy = \sum_{n\ge1}(-1)^{n+1}/((2n)!(2n-2-s))$, while $\int_1^\infty (1-\cos y)y^{-s-3}dy$ and $H^\sharp(s)$ are holomorphic on $\operatorname{Re} s > -2$ because $1-\cos y$ and $\mathcal{K}^\sharp$ are bounded. Let $-2 < \operatorname{Re} s < 0$. The substitutions $y = 2\pi u$ and $y = 4\pi u$ give

$$H^\sharp(s) + \widetilde{H}^\sharp(s) = \int_0^\infty \mathcal{K}^\sharp(u)\,u^{-s-3}\,du = \big((2\pi)^{s+2} + 2^{-11/4}(4\pi)^{s+2}\big)\mathcal{C}(s) = (2\pi)^{s+2}\big(1+2^{s-3/4}\big)\mathcal{C}(s), \tag{16.15}$$

and multiplication by $-\frac{(s+1)(s+2)}{4\pi^2}$ gives (16.14) there. Both sides of (16.14) are meromorphic on the connected set $\operatorname{Re} s > -2$, so the identity holds on all of it. At $s_k = \frac{3}{4} + i(2k+1)\pi/\log 2$ we have $2^{s_k - 3/4} = e^{i(2k+1)\pi} = -1$, and s_k is not real, so it is not a pole of $\mathcal{C}$ or of $A^\sharp$. Hence the right side of (16.14) vanishes at s_k. □

17. The difference $G - A$ and the Riemann zeta function

In this section, and only here and in Section 14, ζ denotes the Riemann zeta function, and Γ denotes Euler's gamma function. We use three cited results. The functional equation in the form [4, (2.1.8), p. 16]

$$\zeta(1-s) = 2^{1-s}\pi^{-s}\cos\Big(\frac{\pi s}{2}\Big)\Gamma(s)\zeta(s). \tag{17.1}$$

The evaluation $\int_0^\infty y^{-w-1}\sin y\,dy = -\Gamma(-w)\sin(\frac{\pi w}{2})$ for $-1 < \operatorname{Re} w < 0$, which is the last step of the first proof of the functional equation in [4, p. 15]. The relations $\Gamma(z+1) = z\Gamma(z)$ and $\Gamma(z)\Gamma(1-z) = \pi/\sin(\pi z)$ for non-integral z, and the fact that $1/\Gamma$ is an entire function [3, §4.41, p. 149].

Reviewed by Claude (review 1)

Proposition 17.1 (The difference $G - A$ through ζ)**.** *For $-2 < \operatorname{Re} s < 0$,*

$$H(s) + \widetilde{H}(s) = \int_0^\infty \mathcal{K}(u)\,u^{-s-3}\,du = (2\pi)^{s+2}\zeta(1-s)\,\mathcal{C}(s), \tag{17.2}$$

and on $\mathbb{C}$, as an identity between meromorphic functions,

$$G(s) - A(s) = \frac{\pi}{s}\cot\Big(\frac{\pi s}{2}\Big)\zeta(s). \tag{17.3}$$

Proof. Let $-2 < \sigma < 0$. The terms of (4.1) are nonnegative, and the substitution $y = 2\pi m u$ gives $\int_0^\infty (1-\cos(2\pi m u))u^{-\sigma-3}du = (2\pi m)^{\sigma+2}\mathcal{C}(\sigma)$. Since $\sum_{m\ge1} m^{-3}(2\pi m)^{\sigma+2} = (2\pi)^{\sigma+2}\sum_{m\ge1} m^{\sigma-1} < \infty$, the theorems of Tonelli and Fubini allow termwise integration for complex s with $\operatorname{Re} s = \sigma$, and $\sum_{m\ge1} m^{s-1} = \zeta(1-s)$. This gives (17.2), where H and $\widetilde{H}$ are as in (5.1) and (14.4). Now let $-2 < \sigma < -1$. Integration by parts in (16.13), whose boundary terms vanish because $(1-\cos y)y^{-\sigma-2}$ tends to 0 at 0 and at ∞, and the evaluation quoted above with $w = s+1$ give

$$\mathcal{C}(s) = \frac{1}{s+2}\int_0^\infty y^{-s-2}\sin y\,dy = -\frac{\Gamma(-s-1)}{s+2}\sin\Big(\frac{\pi(s+1)}{2}\Big) = -\frac{\Gamma(-s-1)}{s+2}\cos\Big(\frac{\pi s}{2}\Big). \tag{17.4}$$

By (5.2), valid on $\mathbb{C}$ by Theorem 5.2, and by (14.4), $G - A = -\frac{(s+1)(s+2)}{4\pi^2}(H + \widetilde{H})$ for $-2 < \sigma < 0$. With (17.2) and $(s+1)\Gamma(-s-1) = -\Gamma(-s)$ this gives, for $-2 < \sigma < -1$,

$$G(s) - A(s) = -(2\pi)^s\,\Gamma(-s)\cos\Big(\frac{\pi s}{2}\Big)\zeta(1-s) = -2\,\Gamma(s)\Gamma(-s)\cos^2\Big(\frac{\pi s}{2}\Big)\zeta(s), \tag{17.5}$$

by (17.1). Since $\Gamma(1-s) = -s\Gamma(-s)$, we have $\Gamma(s)\Gamma(-s) = -\pi/(s\sin(\pi s))$, and $\sin(\pi s) = 2\sin(\frac{\pi s}{2})\cos(\frac{\pi s}{2})$ gives (17.3) for $-2 < \sigma < -1$. Both sides of (17.3) are meromorphic on $\mathbb{C}$, so the identity holds on $\mathbb{C}$. □

Reviewed by Claude (review 1)

Corollary 17.2 (The condition on G and A is the Riemann hypothesis)**.** *For every non-real s, $G(s) = A(s)$ if and only if $\zeta(s) = 0$, and the multiplicity of s as a zero of $G - A$ equals its multiplicity as a zero of ζ. Hence the following two statements are equivalent.* (a) *Every s with $\operatorname{Im} s \ne 0$ and $G(s) = A(s)$ has the form $s = \frac{1}{2} + it$ with real $t \ne 0$.* (b) *Every zero of ζ that is not real has real part $\frac{1}{2}$, that is, the Riemann hypothesis.*

Proof. Let s be non-real. The poles of A are real by Proposition 14.1, so $G(s) - A(s)$ is finite. The function $\frac{\pi}{s}\cot(\frac{\pi s}{2})$ is holomorphic and nonzero at s, because $\sin z$ vanishes only at the real points $z \in \pi\mathbb{Z}$ and $\cos z$ only at the real points $z \in \frac{\pi}{2} + \pi\mathbb{Z}$. Hence (17.3) gives the first statement, with multiplicities, and the equivalence of (a) and (b) follows. □

Through Corollary 17.2, every theorem on the non-real zeros of ζ is a theorem on the non-real solutions of $G = A$. For instance, $\zeta(1+it) \neq 0$ for real $t \neq 0$ [4, §3.1, p. 45], so $G(1+it) \neq A(1+it)$ for $t \neq 0$, and also $G(1) \neq A(1)$ by Proposition 15.1. Let $N(T)$ be the number of zeros of ζ with $0 \leq \operatorname{Re} s \leq 1$ and $0 < \operatorname{Im} s \leq T$, counted with multiplicity [4, §9.2, p. 211]. By Corollary 17.2, $N(T)$ is also the number of solutions of $G = A$ in the same region, and by [4, Theorem 9.4, p. 214]

$$N(T) = \frac{T}{2\pi} \log \frac{T}{2\pi e} + O(\log T) \qquad (T \to \infty). \tag{17.6}$$

Put $N_0(T) = \frac{T}{2\pi} \log \frac{T}{2\pi e} + \frac{7}{8}$, the main term of (17.6) plus the constant $\frac{7}{8}$. On $T > 2\pi$ the function N_0 increases from $-\frac{1}{8}$ to ∞. For every integer $k \geq 1$ let $\tau_k > 2\pi$ be the solution of $N_0(\tau_k) = k - \frac{1}{2}$. This equation reads $\frac{\tau_k}{2\pi e} \log \frac{\tau_k}{2\pi e} = (k - \frac{11}{8})/e$ with $\log \frac{\tau_k}{2\pi e} > -1$, so that $\log \frac{\tau_k}{2\pi e} = W((k - \frac{11}{8})/e)$ and

$$\tau_k = \frac{2\pi(k - \frac{11}{8})}{W((k - \frac{11}{8})/e)}, \tag{17.7}$$

where W is the inverse function of $w \mapsto we^w$ on $[-1, \infty)$. The first values are $\tau_1 = 14.5213$, $\tau_2 = 20.6557$, $\tau_3 = 25.4927$, $\tau_4 = 29.7394$, $\tau_5 = 33.6245$ and $\tau_6 = 37.2574$. The imaginary parts of the six solutions of $G = A$ shown in Figure 8, which are those of the first six non-real zeros of ζ in the upper half-plane, differ from $\tau_1, \ldots, \tau_6$ by at most 0.69. At the points $\frac{1}{2} + i\tau_k$ with $k \leq 6$ the computed values of $|G - A|$ lie between 0.054 and 0.105, so these points are not solutions of $G = A$. The statements of the last two sentences are numerical observations.

18. Conclusions

Sections 15 and 16 characterize the difference $G - A$ without the Riemann zeta function, using only the function h. On the strip $0 < \operatorname{Re} s < 2$ it is minus the Mellin transform (15.2) of the positive function $\ell(u) = \log(\pi u / |\sin(\pi u)|)$, so that $|G - A|$ is bounded by its value $A(\sigma) - G(\sigma)$ on the real axis, which is attained only there. It is also the sum (15.7) of an explicit digamma term, whose real part is negative, and of two transforms of nonnegative remainders. After the factor $-(s+1)(s+2)/(4\pi^2)$, it is the Mellin transform Ω of the convex kernel $\mathcal{B}$ of (16.1). This kernel is positive and continuously differentiable, and on $(0, 1)$ it is a power series with positive coefficients and exponents at least 4. It satisfies the strict inversion inequality (16.6), so that the real and imaginary parts of Ω are cosine and sine transforms of two positive weights. It follows that Ω is positive and decreasing on the real segment $(0, 1]$, that there is the region (16.9) without solutions of $G = A$ around the real axis, and that $\Omega(\sigma + it) = O(t^{-2})$. Theorem 15.4 proves that $A(s) \neq 0$ for $0 \leq \operatorname{Re} s \leq 1$ and $|\operatorname{Im} s| \geq 9$, so that there the equation $G = A$ is the equation $G/A = 1$. The condition on G and A that every solution of $G(s) = A(s)$ with $\operatorname{Im} s \neq 0$ has the form $s = \frac{1}{2} + it$ is a statement about these two functions alone, and it can be formulated without ζ.

None of these properties places the solutions on the line $\operatorname{Re} s = \frac{1}{2}$. The inversion inequality holds for every $\sigma \in (0, 1]$ and does not single out $\sigma = \frac{1}{2}$. Proposition 16.5 gives a kernel $\mathcal{K}^\sharp$ with two harmonics that is nonnegative, has period 1 and vanishes to second order at the integers, exactly like $\mathcal{K}$, and for which the counterpart of $G - A$ vanishes at points with real part $\frac{3}{4}$. Therefore a proof of the condition has to use the coefficients m^{-3} of all the harmonics of $\mathcal{K}$, and not only the positivity and the periodicity of the kernel.

Proposition 17.1 shows where these coefficients go. The sum over all the harmonics of $\mathcal{K}$ is the Dirichlet series of $\zeta(1 - s)$, and $G - A$ equals $\frac{\pi}{s} \cot(\frac{\pi s}{2}) \zeta(s)$. This is the problem that appears when ζ is introduced. By Corollary 17.2 the condition on G and A is then equivalent to the Riemann hypothesis. The introduction of ζ therefore does not reduce the condition to a weaker statement. The step that remains after it is the Riemann hypothesis itself. What ζ brings are the known theorems on its zeros, which become theorems on the solutions of $G = A$. The theorem of Hadamard and de la Vallée Poussin excludes solutions on the line $\operatorname{Re} s = 1$, which the methods

of Sections 15 and 16 do not do. The Riemann–von Mangoldt formula (17.6) counts the solutions with $0 < \operatorname{Im} s \leq T$, and the points τ_k of (17.7) are defined from its main term. But (17.6) and the points τ_k concern only the imaginary parts of the solutions. They give no information on the real parts, and the points $\frac{1}{2} + i\tau_k$ are not themselves solutions. The two routes therefore stop at the same place, the real parts of the solutions of $G = A$ in the strip $0 < \operatorname{Re} s < 1$. Without ζ, the obstruction is that the structure proved so far is shared by kernels whose solutions lie off the line. With ζ, the obstruction is the Riemann hypothesis.

It remains to say whether G and A intersect, and where. They do intersect, and this is not impossible to prove. On the real axis, $G(-1) = 0$ by (5.6) and $A(-1) = 0$ by Proposition 14.1, so $G(-1) = A(-1)$ without any use of ζ, while $G(s) \neq A(s)$ for $0 < s < 2$ by Proposition 15.1. By (17.3), $G(s) = A(s)$ also at every odd integer $s \geq 3$ and $s \leq -3$, because $\cot(\frac{\pi s}{2}) = 0$ there and ζ is finite at every $s \neq 1$ [4, Theorem 2.1, p. 13]. Off the real axis, Corollary 17.2 and Hardy's theorem that ζ has infinitely many zeros on the line $\operatorname{Re} s = \frac{1}{2}$ [4, §10.2, p. 256] show that $G = A$ at infinitely many points $\frac{1}{2} + it$. So it is proved that G and A intersect infinitely often off the real axis, and on the line $\operatorname{Re} s = \frac{1}{2}$ itself. Hardy's proof uses that $\zeta(\frac{1}{2} + it)$ times an explicit factor is real, which is a consequence of the functional equation. The representations of Sections 15 and 16 do not show a symmetry of $G - A$ under $s \mapsto 1 - s$, and this note does not prove the existence of a non-real solution of $G = A$ without ζ. What is not proved is that G and A intersect off the real axis only on the line $\operatorname{Re} s = \frac{1}{2}$. By Corollary 17.2 this is the Riemann hypothesis. It is not known to be impossible to prove; it is open.

References

[1] F. Johansson, Arb: efficient arbitrary-precision midpoint-radius interval arithmetic, *IEEE Trans. Comput.* **66** (2017), no. 8, 1281–1292. doi:10.1109/TC.2017.2690633. (Cited on pp. 11 and 31.)
[2] H. J. S. Smith, On the value of a certain arithmetical determinant, *Proc. London Math. Soc.* **7** (1875/76), 208–212. doi:10.1112/plms/s1-7.1.208. (Cited on p. 2.)
[3] E. C. Titchmarsh, *The Theory of Functions*, 2nd ed., Oxford University Press, Oxford, 1939. No DOI. (Cited on p. 34.)
[4] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., revised by D. R. Heath-Brown, Clarendon Press, Oxford, 1986. No DOI. (Cited on pp. 34, 35, and 36.)

Statements reviewed by Claude: 37.